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Multiview 3D Reconstruction and Facial Landmarks Detection in CMU Panoptic

Master's Thesis Report

Master in Computer Vision

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Abstract

With the advancement of new technologies, there are increasingly more applications where the creation and analysis of facial models is essential. Both in the field of entertainment, using these systems to reproduce facial reconstructions within interfaces, and in the field of security, for the early detection of incidents, it is an area in continuous development. However, training these neural network models requires the use of large 3D facial datasets. Although other multiview laboratory databases exist, the CMU Panoptic dataset stands out for its great demographic and motion diversity.

The main objective of this project is, therefore, to annotate the Panoptic dataset in order to obtain the complete 3D reconstruction of the subjects' heads using multiview information. At each time instant of a Panoptic scene, detections of the landmarks are obtained with the DAD-3DNet model on the images from the different cameras. To select the camera views with the most accurate predictions, several methods are studied based on ranking predictions according to their reprojection error against the Panoptic annotations and a manually created ground-truth. After selecting a method based on pairwise camera comparison using RANSAC, the new 3D reconstructions and the points projected onto the previous images are obtained, forming the new Panoptic dataset.

With this data, the results are validated by using them in a facial analysis task. For this purpose, a facial landmarks detector was selected and two training procedures were carried out with different loss functions. The first procedure assigns the same value in the error to all characteristic points, while the second takes into account the visibility condition of each point. This visibility is extracted according to the subject's orientation in the image and certain geometric constraints. The results are evaluated against the detections from DAD-3DNet and the refined annotations, achieving adequate predictions with both models.

In summary, this project successfully achieves the goal of providing a method to annotate the Panoptic dataset, obtaining the 3D model of the faces and the projection of their points in each image. Furthermore, these data are successfully used to train facial landmark detectors. The reprojection error obtained by the final reconstruction method is less than 15 pixels for the evaluated scenes, reducing the error from a few pixels up to almost 400 pixels compared to a randomly ordered reconstruction. These 3D models are also compared against the original predictions and a facial texturing procedure, showing better alignment on the lateral part of the head. Finally, for the landmarks detectors, the visibility feature is implemented in the training, achieving an appropriate trend in point detection.

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Glossary

3DMM 3D Deformable Model. [2](#), [7](#), [9](#), [10](#), [16](#), [58](#)

BA Bundle Adjustment. [2–5](#), [18–20](#), [30–32](#), [39](#), [43](#), [44](#), [55](#)

DLT Direct Linear Transformation. [5](#), [18](#), [19](#), [44](#)

GAN Generative Adversarial Network. [10](#)

LMM Local Deformable Model. [10](#)

MSE Mean Squared Error. [26](#)

PCA Principal Component Analysis. [10](#)

RANSAC RANdom SAmple Consensus. [2–4](#), [17](#), [20](#), [31](#), [32](#), [39–41](#), [43–49](#), [55](#), [56](#)

RMSE Root Mean Squared Error. [23](#), [24](#), [39](#), [52](#)

SfM Structure from Motion. [13](#)

SfS Shape-from-Shading. [9–11](#)

SVD Singular Value Decomposition. [18](#)

Chapter 1

Introduction

In this chapter, the core content of this Master’s Thesis is presented. It begins by providing the context for the motivation behind the chosen project and outlining the problem to be addressed. Next, the general and specific objectives of this research are outlined, along with the scope and limitations that inevitably influence its development. Finally, a general description of the project’s structure is provided.

1.1. Motivation

In recent years, there has been a notable effort in developing computer vision systems aimed at detecting facial key points and generating 3D facial models. Specifically, the most recent initiatives focus on creating methods capable of interpreting the two-dimensional information in an image and extrapolating these points, known as facial landmarks, into three-dimensional reconstructions. Current methods include the development of DECA (*Detailed Expression Capture Animation*) [1], EMOCA (*EMOtion Capture and Animation*) [2], and DAD-3DNet [3], which will be used in this project. These new capabilities are gaining traction in the field of facial analysis, allowing for comprehensive studies based solely on images and increasing both the quantity and quality of available applications.

The advantages offered by facial reconstruction are closely linked to the wide range of areas in which it can be applied. These include applications in biometric systems, security and surveillance, interface control, virtual face-based search, and even medical analysis through facial features [4] [5]. In the entertainment field, it is used in animation, avatar creation, and the generation of realistic facial models for post-production in multimedia content, video games, or virtual reality [6].

However, despite numerous advances in facial detection and reconstruction from images in calibrated camera environments, there is still much progress to be made, particularly regarding the robustness of systems developed in less controlled settings [7]. This is due to the considerable variability present in facial images. Specifically, factors such as lighting, position, occlusions, and individual facial characteristics can be critical to achieving a high-quality reconstruction that closely resembles reality.

Furthermore, the quality of results is directly influenced by the capture method used to extract the data. For instance, applications requiring highly accurate reconstructions may employ multi-camera depth systems [8] or a combination of cameras with LiDAR

sensors [9], thereby increasing data resolution. However, acquiring such data presents limitations due to the complexity and time required to use these tools, which makes it difficult to create large-scale 3D facial model databases. In this context, although datasets are available today, they still have certain limitations. The problem lies in the biases and constraints associated with dataset creation, such as limited variability in age, gender, or scene characteristics. This is the case with the CMU Multi-PIE database [10]. Naturally, this also affects the final evaluation of models and the accuracy levels achieved. Despite these challenges, significant advances have been made in developing models under different strategies. For example, the evolution of [3D Deformable Models \(3DMMs\)](#) [11] over the past 20 years and the development of deep neural networks have improved application performance in terms of accuracy [12].

For this reason, given the ease of acquiring 2D images and the greater visual accuracy provided by 3D facial models, there is growing interest in translating 2D to 3D, offering the opportunity to increase the availability of three-dimensional models [4]. In particular, the scenario addressed in this project involves using video sequences taken from multiple perspectives to perform a complete reconstruction of the head. The goal is to annotate the CMU Panoptic database with the 3D positions of the head points and their reprojections.

On the other hand, facial landmark detection is a crucial step in applications that utilize these facial models. Early attempts at detecting these reference points relied on statistical models that were heavily influenced by environmental conditions. However, thanks to the introduction of neural networks, some of these issues have been overcome, achieving the robustness needed against facial occlusions or lighting changes and ensuring adequate performance even in uncontrolled environments [13]. Therefore, occlusion of certain parts of an image can drastically affect the detection results obtained. To validate the new 3D reconstruction annotations, these data will be used to train two facial landmark detectors. During one of these trainings, the visibility of points in the image will also be considered, examining its influence on the results.

In summary, the main objective of this Master’s thesis is to implement an algorithm to obtain refined 3D reconstructions using information from different camera views, with the ability to use the new point reprojections on the images for training new facial analysis models.

1.2. Problem statement

To perform the 3D reconstruction of the face, the model developed in [14], called DAD-3DNet, is used. This model is capable of extracting the key points of a person’s head from a single image. The images to be annotated come from the Panoptic database [15], which includes a set of scenes with people performing specific movements and social interactions. By applying DAD-3DNet to the Panoptic images, the facial models from the different camera views for each scene are predicted. These data are then refined by selecting the best views using the [RANDOM Sample Consensus \(RANSAC\)](#) algorithm and the [Bundle Adjustment \(BA\)](#) method to obtain the desired final annotations.

Once these refined reconstructions are calculated, the second part of the project involves reprojecting these 3D points onto the corresponding images to train two landmark detection models. For this purpose, one training will use a loss function that considers all

points equally, and a second training will take into account the visibility of each point based on the subject's orientation in the image.

This approach highlights several key aspects. First, an initial filtering of the images from the Panoptic scene viewpoints must be performed for each frame. This requires considering whether the camera is active in the current sequence, the position of the subject relative to that view, and possible occlusions caused by their own gestures or other people in the scene. Similarly, to perform the subsequent refinement of the reconstructions, [BA](#) will be used in the procedure, so it is necessary to understand how to integrate the functioning of this iterative algorithm. To obtain the best possible reconstruction, a comparison of results obtained by several methods based on pairwise reconstructions and [RANSAC](#) is conducted. This allows the cameras to be ordered and prevents the use of those whose predictions by DAD-3DNet may be more erroneous compared to the other views, avoiding their inclusion in the refined reconstruction.

Regarding the training of the facial landmark detector, a simple model will be used for this purpose. The Panoptic scenes are used as input to this neural network along with the 2D key points reprojected from the new annotations. In this case, the goal is to analyze the influence on training when considering the visibility of the points compared to not using this information. To this end, trigonometric methods are applied to determine the visibility condition of the landmarks based on the orientation of the subject's head in the image.

It should be noted that this project aims to present a method for annotating 3D facial reconstructions on the Panoptic database, so that the data can be used in the development of more complex projects in the future. For this reason, the landmark detection application is also implemented as a study of the usefulness of the new annotations in a real case. However, it should be mentioned that the quality of the data inherently depends on the sequences used and the proper functioning of the DAD-3DNet model.

1.3. Objectives

The main objective of this project is to use the predictions provided by the DAD-3DNet model on the Panoptic sequences to obtain the corresponding facial annotations. This procedure aims to improve the detection of the 3D head meshes provided by DAD-3DNet by combining the predictions from different camera views. As a result, the annotations of the reconstructions and the reprojections of these points onto the images are obtained, creating a new facial dataset.

Using the generated data, a second objective is proposed, aimed at validating the annotations through the training of two deep learning models for facial analysis, specifically for facial landmark detection. In this process, information about the visibility of the points in the images is also included.

Therefore, to achieve the established objectives, it is essential to analyze the existing literature to understand current approaches and address the problem efficiently. Additionally, the functioning of the DAD-3DNet model must be understood in order to apply it correctly to the Panoptic images. Furthermore, several methods for selecting the best cameras will be studied, which is a key step in the project due to its direct impact on the quality of the final data. An evaluation will also be established based on comparisons with

Panoptic annotations and a manually created ground-truth, together with a 3D textured visualization. All these procedures for generating refined reconstructions will be integrated into a single process capable of performing the complete annotation of the database, from which a portion will be used for the final training.

More specifically, the tasks associated with achieving the proposed objectives are as follows:

- Study of the existing literature related to 3D facial model databases and the detection and reconstruction of facial key points.
- Development of a method for the automatic selection of cameras for each frame in the Panoptic sequences, performing an initial filtering based on the face positions and avoiding images with total occlusions.
- Application of the DAD-3DNet model to detect the 3D facial meshes on the selected images for each time step in a scene.
- Construction of the 3D head model through initialization with a linear triangular reconstruction and refinement using [BA](#).
- Study of different algorithms based on pairwise comparisons and [RANSAC](#) for the ordering and selection of the cameras with the best predictions for each frame according to the reprojection error.
- Manual annotation of selected scenes for the evaluation of results.
- Evaluation of the reconstructions calculated by the algorithms based on their comparison with Panoptic annotations and manual annotations.
- Development of the complete algorithm integrating the previous steps for the full annotation of the database.
- Visualization of the final reconstructions by texturing the models.
- Extraction of visible and occluded points from the refined 3D reconstructions based on the head position in the images.
- Training of two facial landmark detection models using the reprojections of the refined points, with one loss function treating all points equally and another considering the previously calculated visibility.
- Evaluation of the detections obtained for both models against DAD-3DNet predictions and the annotations.

In summary, achieving these objectives will require a comprehensive approach encompassing the study of the state of the art, as well as the development and evaluation of the proposed algorithms, thereby ensuring the effectiveness and quality of the obtained results.

1.4. Scope and limitations

The detection of facial landmarks and their reconstruction into a 3D model can offer significant advantages when extended to other applications. Due to the nature of the neural network used (DAD-3DNet), the characteristic points are limited to 5023, from which the complete head mesh is generated.

Although the images were captured in a controlled environment and the multi-camera system is calibrated and fixed, there are still variations that can affect the quality of the reconstructions. Notably, occlusions in the Panoptic sequences represent a significant challenge, requiring robust filtering to avoid using these images in the final reconstruction. Additionally, both the ethnic diversity and the number of people present in the videos are key factors influencing detection accuracy.

One aspect that limits the project’s scope is the processing time required to generate a refined reconstruction for each scene. Each video sequence consists of approximately 20,000 frames, and each scene was recorded using a set of 31 HD cameras. To reconstruct a given frame, images from cameras in which the subject’s face is correctly identified must first be filtered, and the face region cropped to predict the landmarks. Next, an initial linear reconstruction is performed using [Direct Linear Transformation \(DLT\)](#), followed by [BA](#) over several iterations to order and select the cameras for the final reconstruction according to the developed comparison algorithm. Therefore, the time required to reconstruct a single multi-view frame is substantial (about 40 seconds), making it impractical to fully re-annotate the entire CMU Panoptic dataset within the timeframe of this Master’s thesis.

In addition to the inherent computational costs of programming and executing each algorithm, the time needed for manual annotation to evaluate the reconstructions using reprojection error across multiple views must also be considered.

After obtaining these reconstructions, their reprojections onto the original images are calculated, and the points are later classified according to visibility for training the new model. Taken together, all these steps result in a computationally expensive process.

As can be seen, meeting the project’s specifications depends on three major factors: the quality of the model used for landmark detection, the time required to perform the tasks, and the available computational resources. Other variables also play a role, including the number of sequences analyzed, the cameras used for each reconstruction, the processing of limited frames, and algorithm optimization.

1.5. Structure of the thesis

As a general outline, a summary of the chapters is presented, highlighting their main points:

Chapter 1 ([Introduction](#)): introduces the thesis, including the motivation, the problem statement, the objectives to be achieved, and the existing scope and limitations.

Chapter 2 ([State of the art](#)): provides information on available 3D facial model datasets and the models that perform these reconstructions using different methods.

Chapter 3 ([Methodology](#)): includes explanations about the dataset and the neural network used, as well as specifications regarding the algorithms and software employed in the reconstructions, training, and evaluations.

Chapter 4 ([Development](#)): presents information on image processing, the programming of algorithms for camera ordering and selection, and scene reconstruction. It also covers the development of landmark visibility extraction and the training of the new models.

Chapter 5 ([Experimental results](#)): shows the results obtained from the camera ordering and scene reconstruction algorithms, the texturing of the refined final reconstructions, and the evaluation of the trained model.

Chapter 6 ([Conclusions](#)): provides a conclusion on the results obtained, constructively addressing the challenges faced and giving a final perspective on the study's quality.

Chapter 7 ([Future work](#)): outlines some potential future steps to improve the project.

Chapter 2

State of the art

This section provides information on some of the most prominent 3D facial reconstruction datasets, detailing their main characteristics. Additionally, certain studies focused on the development of systems for 3D facial model reconstruction are examined.

2.1. 3D facial reconstruction databases

The growing use of deep learning methods in facial geometry reconstruction has driven the development of larger and more detailed databases [16]. This trend is reflected in the literature, where some of the earliest datasets contained a small number of 3D reconstructions with limited variations between subjects and poses, whereas newer databases provide sufficient data to be used by deep neural networks. Below, some of the most relevant databases are described to understand their scope and characteristics:

- **FG3D** [16]: This dataset was created by merging RGB-D data from three different datasets using textures with a **3DMM** and lighting models, enabling accurate facial registration. The datasets included are: FRGC, containing nearly 5,000 3D facial scans; BP4D, extracting 3,376 frames from 328 2D and 3D videos of 41 subjects; and CASIA-3D, consisting of 4,624 scans from 123 people, using only frontal poses. These three datasets make up FG3D with over 200,000 samples. Additionally, the Florence database was used for cross-validation.
- **DAD-3DHeads** [3]: This is the dataset from which the model used in this project was built. It is one of the largest dense 3D databases for facial structure reconstruction to date, providing 3,500 verified landmarks. The faces include a wide variety of orientations, occlusions, expressions, and lighting conditions. Manual annotations were validated by comparison with 3D scanner data (see Figure 1).
- **BU-3DFE** [17]: This dataset consists of 100 subjects aged 18 to 70, 56 % of whom are female. Participants represent six different ethnicities, each displaying varying degrees of prototypical expressions: happiness, disgust, fear, anger, surprise, and sadness. Two different views were captured, yielding 5,000 images and 2,500 3D reconstructions per expression. The dataset was later expanded with BU-4DFE (3D + time) [18], including video sequences at 25 fps with 100 frames per expression. 101 subjects of different backgrounds and genders participated, and each 3D model

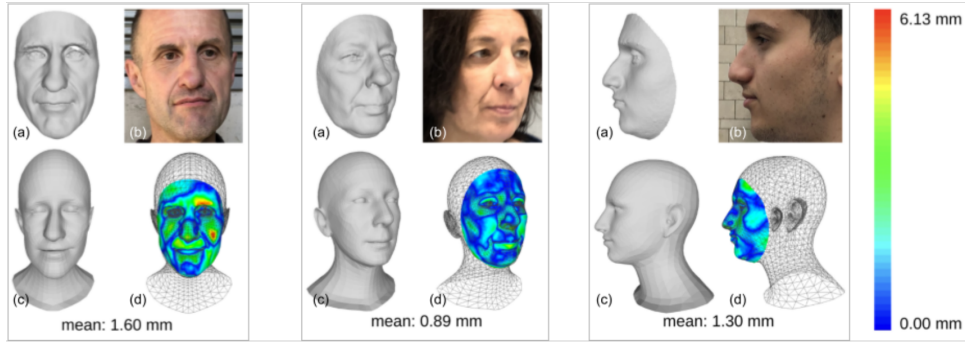


Figure 1: 3D reconstruction error in mm generated from 2D annotations [3].

consisted of 35,000 vertices. The BP4D-Spontaneous database [19] was also added, containing videos of more spontaneous expressions with 41 participants.

- **FaceScape** [20]: This dataset was constructed using 68 densely positioned cameras under controlled lighting, capturing detailed facial features. Data collection involved 938 individuals aged 16–70 performing 20 different expressions, resulting in a total of 18,760 high-quality 3D facial models. Figure 2 illustrates the procedure for capturing and transforming data into 3D models.

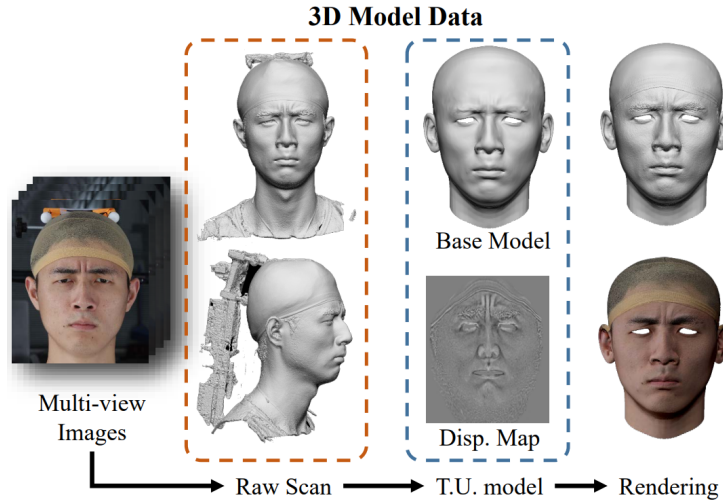


Figure 2: Image capture and transformation process to uniform 3D models [20].

- **DFAW-Video** [21]: This dataset consists of video sequences of 66 subjects aged 18–28. Recordings were captured with iPhones, while the 3D mesh ground truth was generated using high-resolution 3D stereo systems, providing dense reconstructions for each frame.

Overall, there is a clear trend toward creating larger and more detailed databases, focusing on improving models by increasing the available data. Reference data for 3D reconstructions can be generated through manual annotations, 3D scans, or RGB-D cameras. The recurrent use of existing datasets as a foundation to expand or create new datasets with greater diversity and scale is particularly notable.

2.2. 3D face reconstruction

The field of facial analysis, tracking, recognition, and reconstruction has seen significant research development within computer vision [6]. As previously mentioned, such reconstruction can be used in a wide range of applications, from security and access control to planning cosmetic surgeries.

Although prior 2D analysis of images may be necessary for some of these procedures, the use of depth information is highly beneficial for obtaining robust and high-quality systems. However, as noted in [22], acquiring this information can be limited, which has driven growing interest in developing models capable of predicting a 3D facial mesh from a single image, avoiding the need for real-time 3D scanners. These scanners, although widely used in other applications, can pose obstacles due to requirements for proper operation, such as controlled lighting, limitations in automated use, associated costs, or potential risks related to using a laser near the eyes.

According to the approaches employed in studies, [23] classifies 3D face reconstruction methods into several groups, though combinations of these methods are also found in the literature. The methods are as follows:

- Example-based reconstruction methods, where a reference head model is deformed.
- Reconstruction of facial geometry from detected 2D facial landmarks.
- Using techniques such as [Shape-from-Shading \(SfS\)](#).
- Using statistical methods, notably parameter fitting with [3D Morphable Models \(3DMM\)](#).
- Deep neural networks for estimating 3D facial details, including networks capable of producing facial reconstructions from one or multiple images of a person.

For example-based reconstruction, the work presented in [24] is capable of estimating 3D faces while handling some variability in facial appearance. The approach involves an optimization process aimed at maximizing the similarity between facial appearance and depth based on a reference facial model.

Focusing on the use of 2D facial landmarks for 3D reconstruction, the study in [25] has inspired this project. Specifically, it contributed to both refining reconstructions and using landmark visibility in training the subsequent neural network. This work develops a customized optimization based on generating spatially and temporally consistent 3D facial meshes from landmarks extracted from multi-view image sequences. To achieve this, they propose a three-part loss function: the first based on the 2D landmark error, the second on temporal reconstruction error across the sequence, and the third on multi-view consistency that considers images from multiple cameras simultaneously. The last component weights visible landmarks higher, increasing their contribution to model optimization (see Figure 3).

Among the methods mentioned, [SfS](#), developed by [26], is generally used as a post-refinement for an already reconstructed model. However, its use may be less suitable in some cases, as described in [27], because it assumes a direct correspondence between

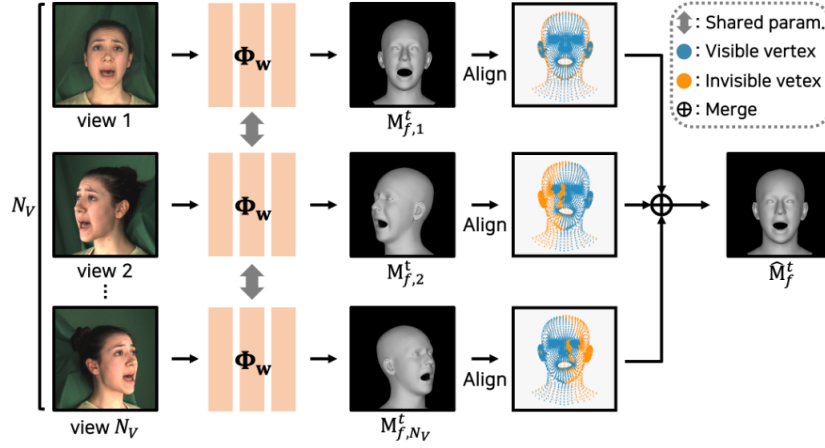


Figure 3: Alignment of multi-view 3D models according to confidence of visible and occluded points [25].

image texture and geometry. Nevertheless, it is widely used in literature, as in works by [28], [29], or combined with other methods such as [Local Deformable Models \(LMMs\)](#) [30], helping to overcome some limitations.

Unlike [SfS](#), [3DMMs](#), introduced by [11], rely on parametric models typically built via [Principal Component Analysis \(PCA\)](#) on a dataset and can estimate and reduce the error between a 2D image and the reconstruction. [3DMMs](#) are also widely used, with examples in studies such as [31] or [32]. They handle varying lighting and head orientations better, though issues arise when facial landmarks are not fully visible.

In this Master’s thesis, the DAD-3DNet model from [3] is used. It predicts the locations of facial landmarks and extracts parameters for head shape, expression, and orientation, generating a complete 3D reconstruction. Its performance is state-of-the-art, trained across a variety of orientations, occlusions, and scenarios. Figure 4 shows how the 3D meshes generated from facial landmarks adapt to the head shapes of different individuals.

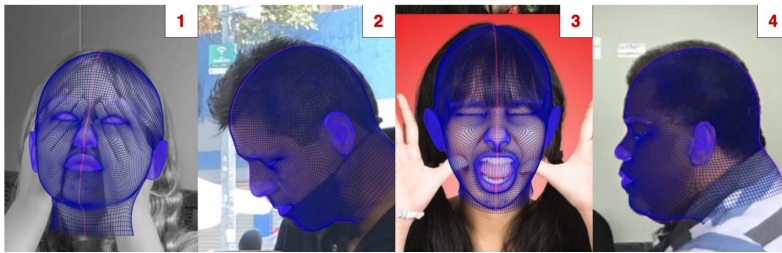


Figure 4: 3D meshes generated by DAD-3DNet [3].

Another project producing 3D facial meshes from a single image is presented in [33], offering a solution using various [Generative Adversarial Network \(GAN\)](#) architectures and comparing results with state-of-the-art methods: autoencoders and [SfS](#) algorithms. A similar combination is seen in [34], whose results are shown in Figure 5. Unlike other works, their training uses synthetic data to produce a [3DMM](#) capable of global reconstruction without intermediate landmarks. Finally, [SfS](#) is applied to refine fine facial features (see Figure 6).

For reconstructions from multi-camera calibrated images, stereo-pair systems can estimate refined disparity maps. These systems are useful when reconstructing a scene from different



Figure 5: Textured reconstructions from [34].

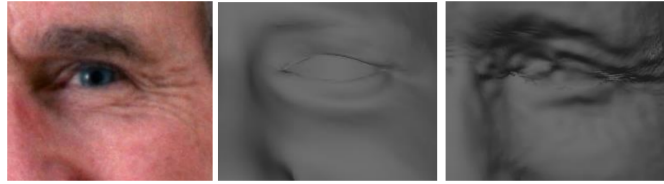


Figure 6: Feature refinement using Sfs [34].

perspectives by using image correspondences and depth information. [29] employs Sfs refinement after generating reconstructions with stereo cameras. This method is applied not only to faces but also to objects under various lighting, sometimes achieving geometric detail superior to laser scanners (see Figure 7).

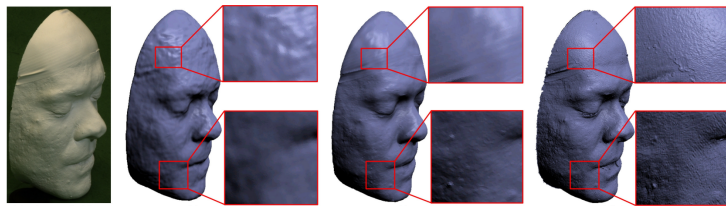


Figure 7: Comparison of facial reconstructions obtained with different methods [29].

Additionally, [7] developed an approach separating images of the same person by head orientation. Most of these images, obtained online, are uncalibrated and used for dense 3D reconstruction. They also developed an algorithm to obtain lateral views of occluded regions with depth constraints. Similarly, [35] uses online images of the same person in different poses, reconstructing iteratively over a 3D model (see Figure 8).

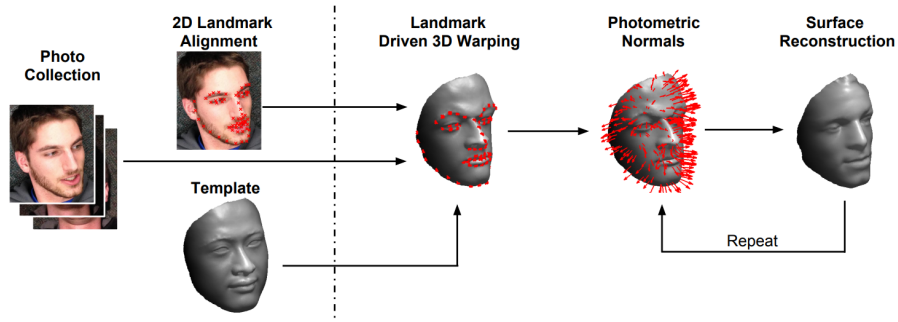


Figure 8: Iterative process for creating a facial model [35].

In summary, many approaches exist for obtaining realistic facial reconstructions. The challenges faced by researchers and the corresponding techniques developed have been discussed. From traditional methods to modern neural networks, this field continues to advance, significantly improving the quality of applications using 3D facial systems.

Chapter 3

Methodology

In the following sections, an analysis of the tools and methodology used is carried out. A detailed description of the Panoptic dataset and the DAD-3DNet model is provided, as well as the reconstruction and refinement process. In addition, the method used to evaluate and visualize the reconstructions is presented. Finally, the training strategy using the new annotations is described, along with a description of the extraction of the visibility of facial reference points.

3.1. Analysis of the Panoptic dataset

Since the main focus of this work is the annotation of the Panoptic scene set, an analysis of its characteristics will be conducted. This dataset is publicly available and was introduced in [15], where both the development carried out for its construction and the functionalities offered are detailed.

The primary reason for its creation lies in the visual detection of human motion structure during certain interactions without the need to use markers. In this way, the aim was to build a large visual dataset to be used for the analysis of body poses and social interactions. Thanks to the study developed in [15], the poses of people in each frame were automatically annotated, overcoming occlusions present in the scenes thanks to the multiple viewpoints. Following this, other projects such as [36], [37], and [38] have enriched the dataset, increasing the analyses and the content of the data.

3.1.1. Multicamera structure

For the purpose of capturing the scenes, the structure shown in Figure 9 was built, covering a diameter and height of 5.49 and 4.15 meters, respectively. It is composed of pentagonal and hexagonal panels, with a set of 24 VGA cameras per panel, 31 HD cameras, 5 projectors, and 10 Kinects with RGB+D sensors throughout the setup, totaling 480 cameras. In the case of this work, only the videos captured by the HD cameras were used, which have a resolution of 1920x1080 and were calibrated using [Structure from Motion \(SfM\)](#). Additionally, they were synchronized with a central clock to ensure that recordings were made at the same frequency.

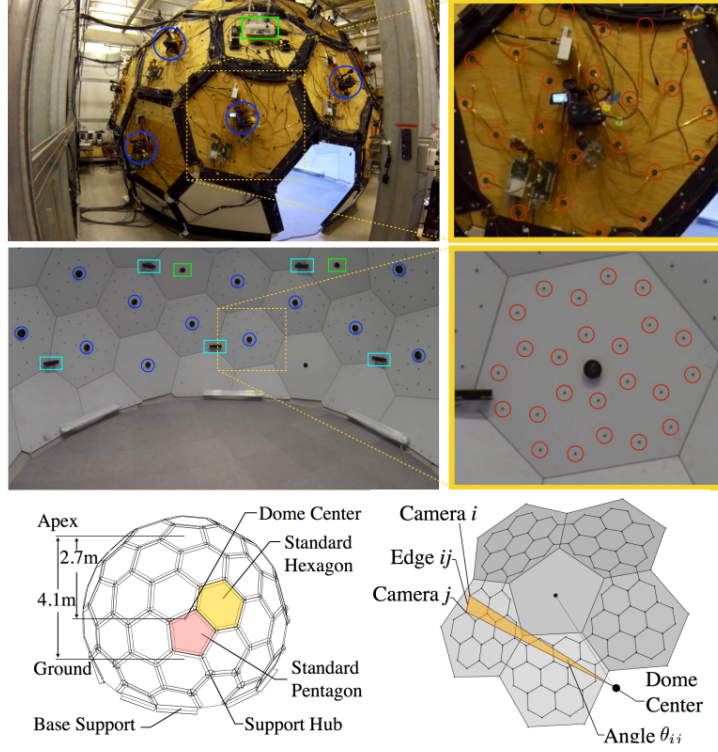


Figure 9: Structure for recording Panoptic scenes [15].

3.1.2. Available annotations

Panoptic provides a large amount of annotations for each image captured by the cameras. Specifically, the dataset offers the key points of each person in every frame, marking the facial and body landmarks, as well as the positions of the hands and fingers with identifiers in each case (see Figure 10).

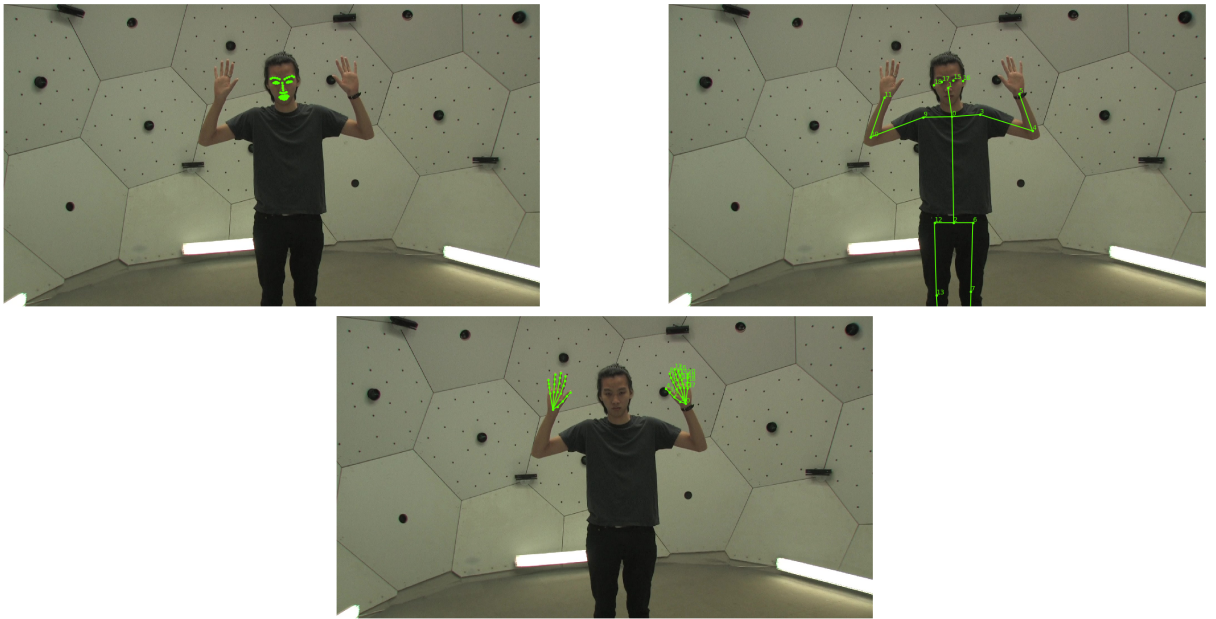


Figure 10: Annotations provided by Panoptic images.

De entre todas las anotaciones, se utilizan los puntos mostrados en la Figura 11 para el preprocesamiento de las imágenes. Estos sirven como referencia para evitar las situaciones de oclusiones entre la misma persona y con otras. En blanco se muestran los puntos de las manos, en verde los puntos que serán usados para detectar la posición del sujeto y otras oclusiones y en rosa el punto correspondiente a la posición de la boca.



Figure 11: Selected Panoptic points used for preprocessing.

Likewise, 27 characteristic points of the face are selected (Figure 12), which will be used in the development of the reconstruction algorithm and the subsequent evaluation of the results.



Figure 12: Selected Panoptic points used in the evaluation.

However, it should be noted that some of these annotations may not be entirely correct. For example, Figure 13 shows some cases comparing the Panoptic points (in red) with manually annotated points for this Master's Thesis (in white).

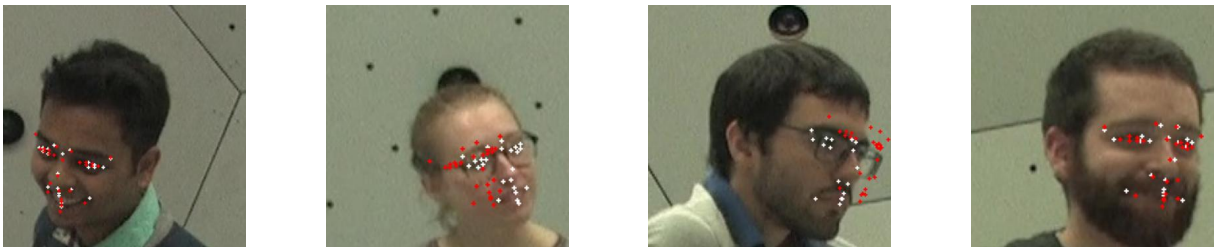


Figure 13: Comparison between the points annotated by Panoptic and those annotated manually.

It can be observed how the positions of the landmarks differ significantly in certain images, so this potential error must be taken into account, as it will affect the final quality of the results.

3.2. DAD-3DNet model

The DAD-3DNet neural network [14] will be used to perform the predictions of the facial landmarks that will be employed for the new 3D reconstructions. It was trained with the DAD-3DHeads dataset using images of size 256x256 pixels. The architecture is shown in Figure 14 and is composed of the elements described below:

- A convolutional encoder to extract the image features.
- A facial landmarks estimator with heatmaps.
- A module to fuse the two previous components.
- A regression module to refine the position of the landmarks and the FLAME [39] parameters, a 3DMM face model.
- A differentiable FLAME [39] layer that generates the 3D face points from the estimated 3DMM parameters.

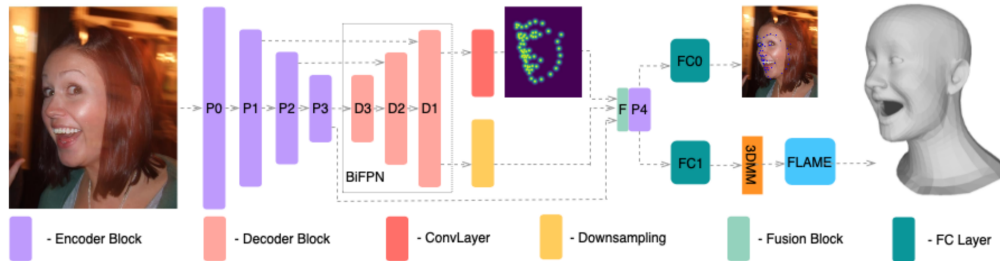


Figure 14: Structure of the DAD-3DNet model [14].

This model, implemented in PyTorch and initialized with ImageNet weights, uses a loss function divided into several parts: the shape and expression loss, the reprojection error, and the Gaussian loss of the heatmap. These calculated errors are responsible for adjusting the head geometry, its orientation, and the position of the landmarks. On the other hand, DAD-3DNet was evaluated by calculating the reprojection error of the 3D mesh obtained with FLAME in each image, in order to validate the position of certain reference landmarks and the predicted subject orientation. Furthermore, the obtained results were compared with other state-of-the-art methods and evaluated under specific scenarios. After applying DAD-3DNet to an image, the predictions shown in Figure 15 can be obtained. In this project, the 5023 predicted landmarks will be used. Additionally, the estimated head orientation shown in the third image of the second row will be utilized to identify and extract the occlusions of the landmarks in the image. This element will be fundamental in the training process of the new models that will use the refined annotations.

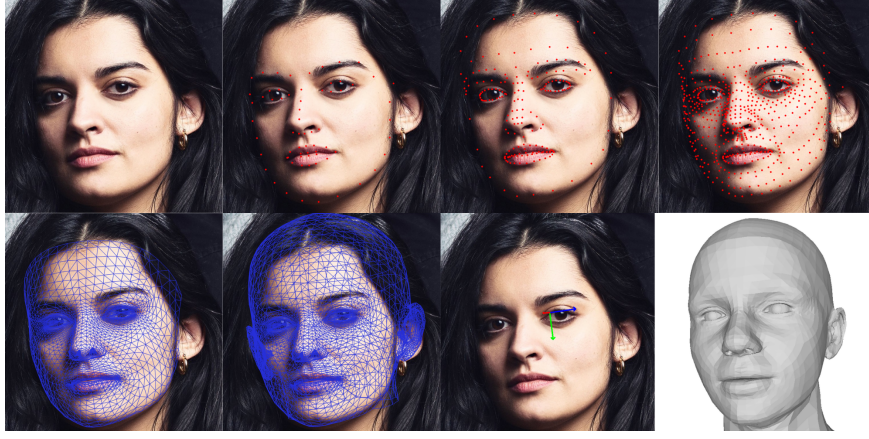


Figure 15: Predictions provided by DAD-3DNet.

3.3. Camera parameters

Camera calibration is essential for projects involving coordinate transformations from 2D to 3D space and vice versa. Specifically, Panoptic provides the intrinsic and extrinsic parameters of each of the 31 HD cameras. These parameters are presented in matrix form, as shown in Equation 1:

$$K = \begin{bmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}, \quad R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}, \quad t = \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix} \quad (1)$$

where K is the intrinsic parameter matrix, and R and t are the extrinsic parameters representing the orientation and translation of the camera with respect to the scene's coordinate system. These matrices can, in turn, be combined to obtain the projection matrix of each camera, as shown in Equation 2.

$$P = K [R \mid t] \quad (2)$$

This 3x4 matrix is useful both for the calculation of the linear reconstruction and for obtaining the reprojections of the 3D points from the reconstruction onto the images.

3.4. Reconstruction methodology

This section presents the developed reconstruction method. DAD-3DNet provides the projection of 5023 3D points onto the images from each of the 31 cameras. These projections are calculated image by image and will be used to perform the 3D reconstruction of the head. However, the projections are sometimes not as accurate as desired (see Figure 25), so it is necessary to determine which cameras are error-free in order to use them for the reconstruction. To carry out this process, several algorithms are developed to compare predictions based on reconstruction errors and the use of RANSAC. These methods are responsible for finding the cameras where the reconstruction projects onto the annotated landmarks and selecting those that will be used to obtain the final 3D coordinates.

For each reconstruction, an initial linear estimation is required, followed by a refinement using [Bundle Adjustment \(BA\)](#). These procedures will be explained below, along with the algorithms developed for ranking and selecting the best predictions.

3.4.1. Linear triangular reconstruction

The linear triangular reconstruction algorithm is based on [DLT](#) (see Equation 4) and is used to obtain an initial approximation of the 3D reconstruction of the head from two views. However, this procedure can be extended to more sets. Following the explanation provided by Zisserman in [40], for each image there are the correspondences $u_i = P_i M$ and $u_j = P_j M$, which represent the transformation of a 3D point, M , to the Cartesian coordinates of each image, where i and j are two different cameras and u_i and u_j are the corresponding projections. This is illustrated more clearly in Equation 3.

$$\lambda \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = P \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} \quad (3)$$

Where:

- x and y are the Cartesian coordinates of the projection of M onto the image.
- λ is the scale factor.
- P is the camera projection matrix.
- $M = [X, Y, Z, 1]^\top$ are the homogeneous coordinates in three-dimensional space.

To obtain the system of equations $AM = 0$, the cross products $u \times PM = 0$ in Equation 4 are first calculated, where the $p_i^\top$ represent the rows of the projection matrix.

$$\begin{aligned} x(p_3^\top M) - (p_1^\top M) &= 0 \\ y(p_3^\top M) - (p_2^\top M) &= 0 \\ x(p_3^\top M) - y(p_1^\top M) &= 0 \end{aligned} \quad (4)$$

Of the three equations, two are linearly independent, and the system matrix, A , can be constructed as shown in Equation 5.

$$A = \begin{bmatrix} xp_{3,i}^\top - p_{1,i}^\top \\ yp_{3,i}^\top - p_{2,i}^\top \\ x'p_{3,j}^\top - p_{1,j}^\top \\ y'p_{3,j}^\top - p_{2,j}^\top \end{bmatrix} \quad (5)$$

Therefore, for each point to be reconstructed in one image and its corresponding point in the other image, the matrix A is constructed and [Singular Value Decomposition \(SVD\)](#), $A = UDV^T$, is performed, obtaining the 3D coordinate value from the last column of V , corresponding to the smallest singular value. By doing this for all point correspondences, the desired reconstruction of the subject's head can be obtained.

3.4.2. Refinement with *Bundle Adjustment*

The **BA** algorithm is typically used to refine a reconstruction through iterative steps that minimize the error between the reprojection of the 3D points and the original landmarks in each image. Given an initial 3D reconstruction, such as the one obtained with **DLT**, this method allows recalculating both the camera projection matrices and the 3D points, thereby reducing this error. Following the description in [41], Figure 16 illustrates **BA**, showing two cameras and the reconstruction of five points. The points u_{ij} correspond to the original landmarks detected in each image, where i is the current landmark and j the camera. In contrast, u'_{ij} represents the reprojections of the 3D points, denoted as M_i , from the reconstruction onto the images.

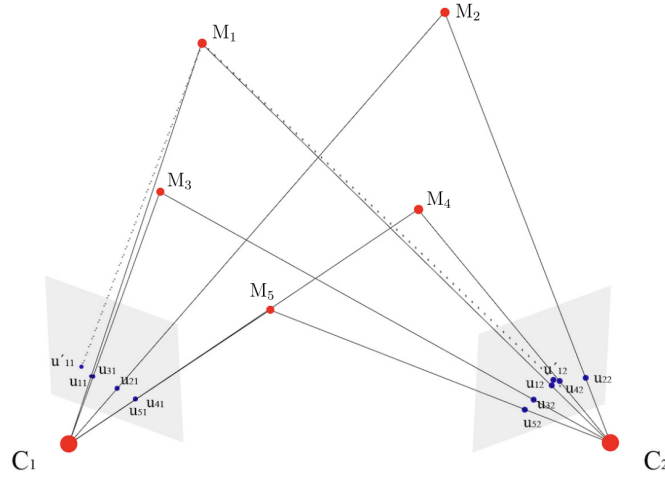


Figure 16: Visual representation of **BA**

Therefore, the minimization is performed over the difference between the key points and their reprojections. In Equation 6, the reprojection error calculation is shown, where n is the number of points, m the number of cameras, and θ the parameters to be optimized.

$$S = \sum_{j=1}^m \sum_{i=1}^n \sqrt{(\vec{u}_{ij} - \vec{u}'_{ij}(\theta))^2} \quad (6)$$

A major advantage of this algorithm is that it can handle missing data in some images. Additionally, its execution can be approached in three different ways depending on the reconstruction specifications:

- **Bundle Adjustment Motion Structure:** assumes that both the calibration and the initial reconstruction can be refined, proposing new parameters and 3D point positions to minimize the error.
- **Bundle Adjustment Motion:** this mode is useful when the 3D points are considered correct and the goal is to estimate the camera calibration from them, so that the error with respect to the scene is minimized. An initial camera calibration must be provided.

- **Bundle Adjustment Structure:** this is the case used in the Master’s Thesis. The camera calibration is assumed to be correct, so it is not modified during optimization. The error exists only in the 3D points, and the coordinates are refined to fit the rest of the images and the projections of their landmarks.

To perform this refinement of the 3D points with [BA](#), the CvSBA package¹

3.4.3. Camera ranking and selection

From now on, we will refer to a *frame or instant reconstruction* to denote the 3D reconstruction obtained by processing the images from the HD cameras. The reconstruction of a face at a given instant is performed using the predictions generated by the DAD-3DNet model for each view, applying the linear procedure and refinement described in the previous sections. In this process, the proper selection of cameras plays a crucial role, as it determines the quality and accuracy of the reconstruction. However, since the DAD-3DNet model makes predictions using a single image, there may be situations where the predictions contain significant errors.

Therefore, the proposed algorithm for performing the final reconstruction must be able to organize the cameras to avoid those views where the error is more pronounced. The operation of this algorithm can be summarized in two objectives:

1. Achieve an approximation of the optimal camera ranking. That is, arrange the cameras so that those producing the lowest reconstruction error appear at the top of the list.
2. Select the number of cameras that results in the lowest reconstruction error.

To address this challenge, three possible algorithmic strategies have been developed, and a study has been conducted on the performance of each to select the most suitable for the task. The study focuses on assessing the impact of camera ranking on the reconstruction compared to a completely random selection of the camera set. The developed algorithms are as follows:

- **Pairwise algorithm with Panoptic:** evaluating the reconstructions from camera pairs against the Panoptic annotations.
- **RANSAC algorithm with Panoptic:** comparing the reconstructions from camera pairs against Panoptic using the [RANSAC](#) implementation. In this way, not all camera pairs are evaluated, reducing the execution time when the number of predictions to compare exceeds seven.
- **RANSAC algorithm with DAD-3DNet predictions:** follows the same procedure as the previous case but evaluates the error against the original DAD-3DNet predictions.
- **Random algorithm:** by randomly ordering the cameras.

The development and operation of these methods will be described in more detail in Section 4.2.3. Likewise, the proposed evaluation method for selecting the best algorithm is presented in Section 3.5, and the results are shown in Section 5.1.

¹<https://www.uco.es/investigacion/grupos/ava/portfolio/cvsba/>

3.5. Evaluation method for the reconstructions

This section presents the method used to evaluate the reconstructions obtained in Panoptic. This evaluation is essential for determining the accuracy and effectiveness of the facial reconstruction algorithms employed in this study. The process is divided into several stages, ranging from the selection of the scenes to be evaluated to the calculation of reprojection errors for each characteristic point. The key phases of this evaluation method are detailed below.

3.5.1. Selection of evaluation scenes

In order to ensure a thorough and diverse analysis, 13 time instants were selected along with the 31 camera views from 6 different videos in the Panoptic database. This evaluation set includes a total of 15 different individuals, with views without occlusions specifically chosen through the filtering process detailed in Section 4.1. In total, a set of 183 images was manually annotated to carry out this task.

The selected scenes are shown in Figure 17 from the perspective of camera 00_00. Variations in poses, features, and types of partial occlusions are observed, such as the use of glasses. It should be noted that, for some of these frames, information from the positions of two different people is used.

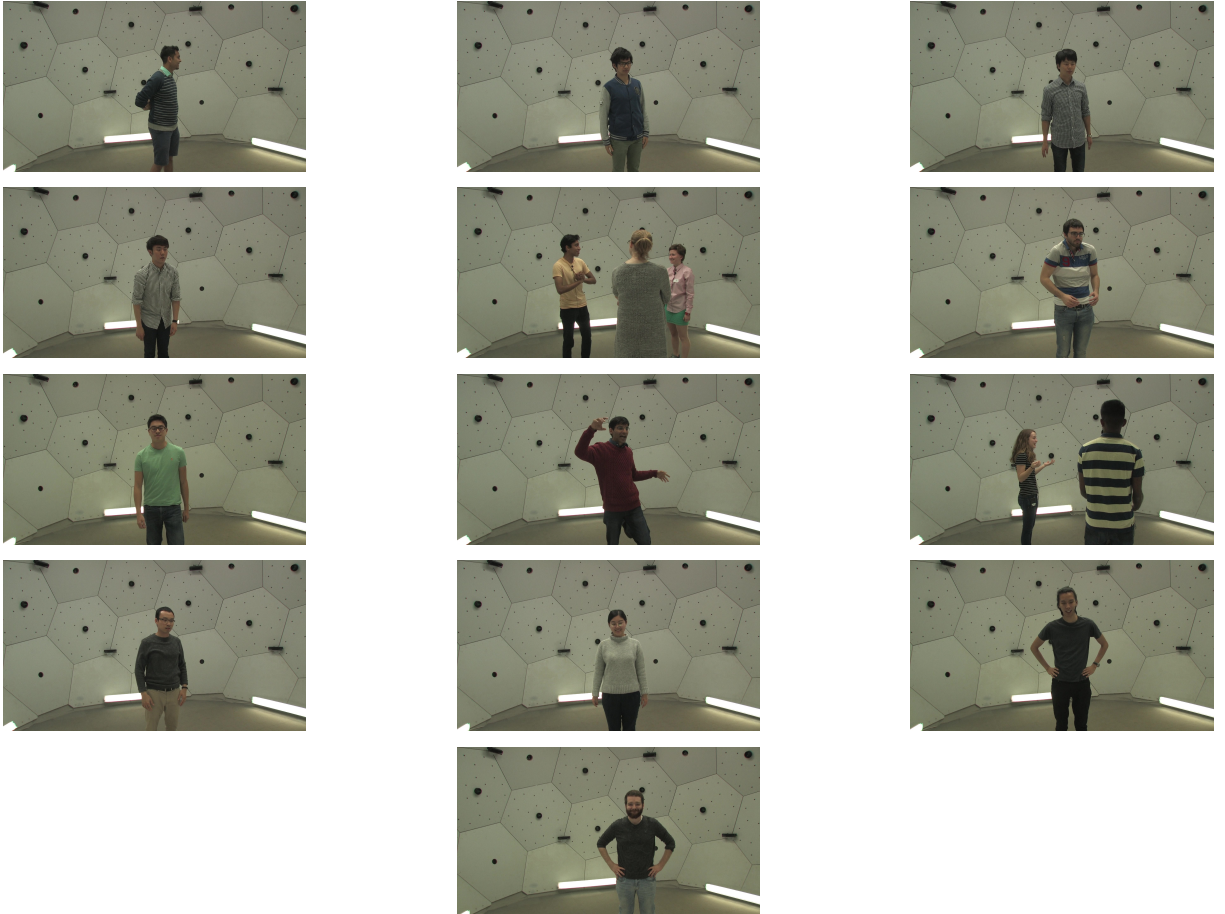


Figure 17: Selected evaluation scenes.

Table 1 provides information about each scene and its identifier, which will later be used to refer to each one. Additionally, the selected frame from each sequence is shown, indicating the subject number used and the total number of filtered views in which the person’s face can be clearly seen.

Table 1: Especificaciones de las escenas de evaluación seleccionadas.

Identifier	Scene	Frame	Subject	Number of views
1	171204_pose5	153	0	13
2	171026_pose1	2008	0	14
3	171026_pose2	2595	0	14
4	171026_pose3	2616	0	12
5	170404_haggling_b3	3270	3	15
6	170404_haggling_b3	3270	4	10
7	171026_pose2	8799	1	10
8	171204_pose5	10808	2	11
9	171026_pose1	14790	1	9
10	170404_haggling_b1	15320	16	15
11	170404_haggling_b1	15320	17	15
12	171026_pose1	16132	2	13
13	171204_pose5	17524	3	11
14	171204_pose5	22633	4	10
15	171204_pose5	26918	5	11
			Total	183

3.5.2. Manual annotation of the images

To accurately evaluate the reconstructions, manual annotation of certain facial landmarks on the previously presented scenes is proposed. For this purpose, the CVAT application²

Among the 5023 facial landmarks, the same 27 points corresponding to those selected from the Panoptic annotations (see Figure 12) are chosen, establishing a common reference. In this way, a common structure is defined with the position of each point, and its location in the image is manually adjusted, as shown in Figure 18.

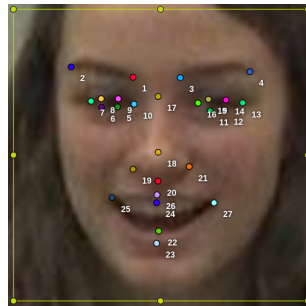


Figure 18: Structure of the 27 landmarks in CVAT.

Manual annotation is carried out for each view of every frame in the evaluation scenes, also taking into account points that are occluded in the image. If any of these points

²<https://www.cvat.ai/>

are not visible, they are marked as such and subsequently excluded from the evaluation calculation.

3.5.3. Evaluation of the refined reconstructions

The evaluation is carried out by measuring the reprojection error of the 3D points from the refined reconstruction. This error is calculated both against the annotations provided by CMU Panoptic and against our manual annotations. This evaluation procedure will be used to select the final reconstruction algorithm and to validate the final reconstructions. It should be noted that, to evaluate the ranking methods, a set of reconstructions is performed for each scene using a different number of cameras, so the average errors are obtained considering this set of reconstructions.

The calculation performed to obtain the reprojection error per pixel is based on the OpenCV function *computeReprojectionErrors*³

1. **landmark error:** the reprojection error calculated for each characteristic point across all views of a scene. To obtain the reprojection error of a landmark in an image, the following Equation 7 is used:

$$\mathcal{E}_l = \|u_l - \hat{u}_l\|_2 \quad (7)$$

where l is the current point for which the error is being calculated in a specific image, with u_l being the refined landmark and $\hat{u}_l$ the ground-truth point. To obtain the reprojection error of this landmark across all views of a scene, Equation 8 is used:

$$\mathcal{E}_{le} = \sqrt{\frac{\sum_{c=1}^C \mathcal{E}_{lc}^2}{L_{le}}} \quad (8)$$

where C is the total number of images and $\mathcal{E}_{lc}$ is the error of a landmark for the current image, c , divided by the total number of times the landmark, l , appears in all images of the scene e (L_{le}). Overall, this gives the **Root Mean Squared Error (RMSE)** of a landmark across all images of a scene. This error calculation will be used in the evaluation of the final reconstruction obtained.

2. **Image error:** provides the reprojection error for a single image ($\mathcal{E}_c$) following Equation 9:

$$\mathcal{E}_c = \sqrt{\frac{\|u_c - \hat{u}_c\|_2^2}{L_c}} \quad (9)$$

where $u_c \in \mathcal{R}^{2L_c \times 1}$ represents the matrix of landmarks to be evaluated for an image, c , and $\hat{u}_c \in \mathcal{R}^{2L_c \times 1}$ represents the landmarks matrix of the image against which the points are evaluated. In this case, L_c is the total number of visible landmarks in the current image.

³<https://github.com/kipr/opencv/blob/master/samples/cpp/calibration.cpp>

3. **Reconstruction error:** to obtain the [RMSE](#) of the reprojection errors of a reconstruction, the error of these points in the images is taken into account. This calculation will be used in the evaluation of the ranking methods and the final reconstructions. It is defined in Equation 10:

$$\mathcal{E}_r = \sqrt{\frac{\sum_{c=1}^C \mathcal{E}_c^2}{L_e}} \quad (10)$$

which maintains the same notation, with C representing the total number of cameras and $\mathcal{E}_c$ being the error calculated for each view, c . This is divided by the total number of visible points across all cameras in a scene (L_e).

4. **Scene error:** shows the mean [RMSE](#) of the landmarks across all reconstructions performed for a ranking method, following Equation 11:

$$\mathcal{E}_e = \frac{1}{R} \sum_{r=1}^R \mathcal{E}_r \quad (11)$$

where $\mathcal{E}_r$ is the error of a specific reconstruction, r , out of the total number of reconstructions R for the scene, e .

5. **Method error:** shows the mean of the errors calculated above, taking into account all the reconstructions performed for all scenes using a specific ranking method. Equation 12 is used:

$$\mathcal{E}_m = \frac{1}{E} \sum_{e=1}^E \mathcal{E}_e \quad (12)$$

where E is the total number of scenes and $\mathcal{E}_e$ represents the error for a scene, e , calculated previously.

These equations will be used in Sections 5.1 and 5.2 for calculating the reprojection error and provide a detailed view of the performance of the reconstruction methods and the final reconstruction produced by the selected method.

3.6. Visualization of the 3D reconstructions

To visualize the final reconstructions, the Python VTK module is used, which can process a triangle-based mesh in real time and display it with texture. To achieve this visualization, an algorithm based on point connectivity is developed to generate triangles. The connectivity information between landmarks used by DAD-3DNet to draw the meshes is employed. When two points are defined as connected, it is checked whether they are also connected to a third point. From these connections, triangles are constructed to visualize the complete textured head.

3.7. Training of 2D landmark detectors with 3D reconstructions in Panoptic

This section presents the methodology for training new landmark detection models using annotations derived from the projection of 3D reconstructions. This approach aims to integrate information about facial occlusions caused by head pose, in order to assess the final accuracy of the predicted 2D points.

3.7.1. Definition of visibility for annotated 2D landmarks

Before starting the training, the visibility of the landmarks is analyzed according to the head orientation. First, this orientation is obtained as predicted by the DAD-3DNet model for a specific image (see Figure 15). To determine whether the landmarks are visible in the image, the 3D reconstruction must be used, so the orientation vector calculated for the image is transferred to the 3D model. In this way, the 3D points can be divided using a plane over the head reconstruction into sets of visible and occluded points. This procedure will be explained in greater detail in Section 4.3.

3.7.2. Training of 2D landmark detectors with the new annotations

The development of the final application with the new annotations is based on the analysis of training a new model with Tensorflow, taking into account the occlusions of the landmarks. The neural network used is the one proposed by Yin Guobing⁴ and is suitable for performing facial landmark detections. The architecture of this neural network consists of six convolutional layers defined by batch normalizations and convolutional filters of different sizes. Furthermore, a pooling operation is performed for each of them. Then, the output from the convolutional layers is flattened and connected to a dense layer of size 1024, activated by ReLU and normalized. Finally, the output layer, also dense, generates the predictions with a linear activation function. This design allows it to learn features from the input images and perform detections of the facial reference points. Since these predictions are based on the position of the landmarks, it is worth mentioning that this is a type of regression problem where the objective is to predict continuous numerical values that define the coordinates of the points.

The input images to the model will be 256x256 pixel crops showing only the subjects' faces. On the other hand, the labels correspond to the 5023 refined points projected onto the image.

The training will be performed twice, changing the loss function used. In this way, the aim is to compare the prediction performance when including information about point visibility and occlusion versus using an equivalent loss for all points. The two loss functions used in each training session are as follows:

⁴<https://github.com/yinguobing/cnn-facial-landmark/tree/master>

1. Using [Mean Squared Error \(MSE\)](#) according to Equation 13:

$$\mathcal{L}_1 = \frac{1}{n} \sum_{i=1}^n (y_{\text{true},i} - y_{\text{pred},i})^2 \quad (13)$$

where n is the total of landmarks, $y_{\text{true},i}$ is the actual position of the landmarks and $y_{\text{pred},i}$ is the position of the landmarks predicted by the model.

2. Also using [MSE](#) but reducing the loss value of the non-visible landmarks by 70 %. To do this, the average of the differences corresponding to the visible and non-visible landmarks is calculated as in Equations 14 and 15:

$$\mathcal{L}_{\mathcal{O}} = 0,3 \cdot \frac{1}{|\mathcal{O}|} \sum_{i \in \mathcal{O}} (y_{\text{true},i} - y_{\text{pred},i})^2 \quad (14)$$

$$\mathcal{L}_{\mathcal{V}} = \frac{1}{|\mathcal{V}|} \sum_{j \in \mathcal{V}} (y_{\text{true},j} - y_{\text{pred},j})^2 \quad (15)$$

where $\mathcal{O}$ is the set of occluded landmarks and $\mathcal{V}$ the set of visible ones. Furthermore, $|\mathcal{O}|$ represents the cardinal of the set $\mathcal{O}$ and $|\mathcal{V}|$ that of the set $\mathcal{V}$. Finally, both values are added together, calculating the final loss value as in Equation 16:

$$\mathcal{L}_2 = (\mathcal{L}_{\mathcal{O}} + \mathcal{L}_{\mathcal{V}}) \quad (16)$$

In this way, the different loss functions dependent on point visibility that will be implemented in the training sessions are defined.

Chapter 4

Development

In this section, each step followed in the development of this project is described in detail. Figure 19 shows a diagram that visually summarizes this structure.

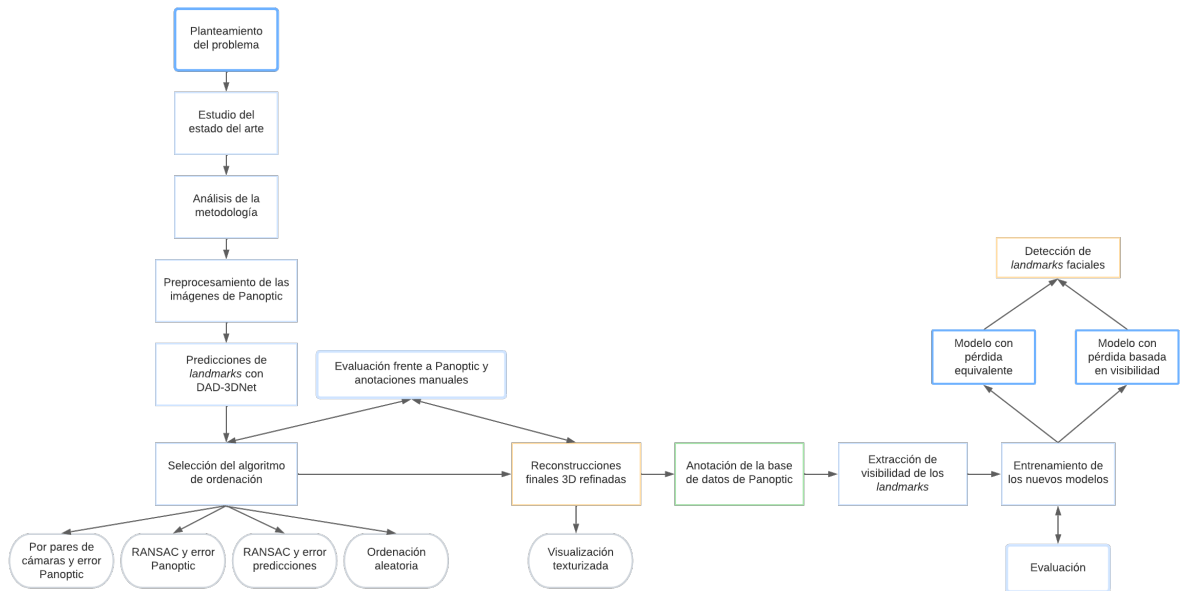


Figure 19: Project diagram.

After studying the proposed approach and its characteristics, the project development begins. This section first covers the preprocessing performed on the Panoptic scene images to filter and crop them so that they can be used with the trained DAD-3DNet model.

Similarly, the algorithms used for the ordering and selection of cameras employed in the final reconstruction are analyzed. Finally, the process for extracting the visibility of the refined points and training the new neural networks for 2D landmark detection is detailed.

4.1. Preprocessing of Panoptic images

To make predictions with DAD-3DNet and process the reconstructions, it is first necessary to obtain the Panoptic scene images in the expected format. For this purpose, the HD

camera images are processed to identify faces and occlusions to obtain the dataset to be used.

Since the DAD-3DNet model works with images where the person’s face occupies nearly the entire frame, Panoptic must be prepared to achieve this type of visualization. This process involves identifying faces in the scene and cropping those areas while considering the size and orientation of the head. Using some annotations provided by the Panoptic data (see Figure 11), an algorithm is implemented that uses the positions of these points to validate the annotations, handle occlusions, and extract the pose of each subject.

If the positions of the characteristic points of the ears, nose, and eyes of the subject appear in order from left to right as expected, the person has a frontal pose. If they appear in the opposite order, the person is facing away from the camera, and the image is discarded since DAD-3DNet requires the face to be at least partially visible. If the nose landmark is positioned farther to the left or right than the other points, the head is considered to be in a profile pose. Finally, the original image crop is adjusted according to the estimated head pose (frontal, left profile, or right profile).

Additionally, certain conditions are applied regarding the distances between the eyes, ears, and nose to prevent excessively pronounced profile poses or situations where the camera is positioned above the head, which could lead to extreme occlusions. Using the chest landmark coordinates, it is possible to verify if the person is looking downward, helping to distinguish between a frontal pose and one in which only the top of the head is visible to the camera. Figure 20 illustrates some positions that would be retained for reconstruction according to these conditions, while Figure 21 shows three examples of camera views that would be filtered (rejected) during this preprocessing.

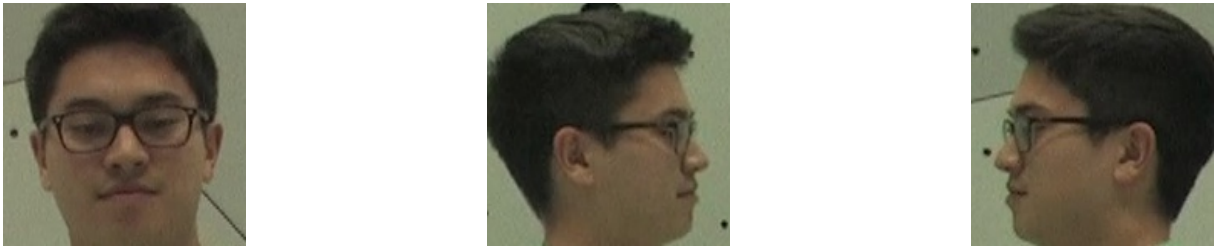


Figure 20: Valid poses.

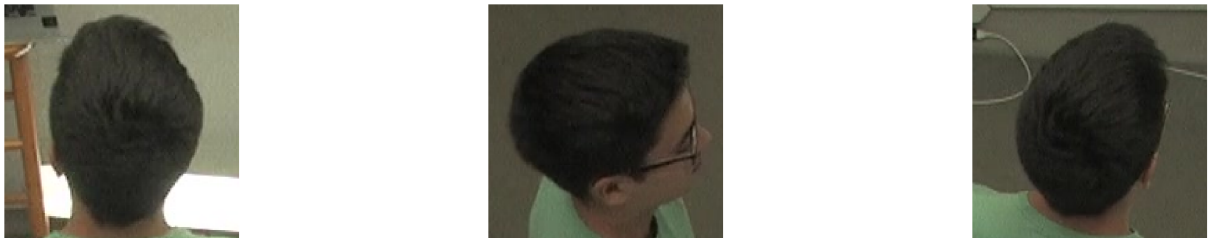


Figure 21: Invalid poses filtered.

In the previous Figures 20 and 21, it can be seen how the person’s pose changes depending on the camera position for the same frame, allowing this filtering to be performed based on the relationship between the points mentioned.

After the initial processing to determine the person’s pose, it is crucial to consider possible occlusions, whether caused by the individual themselves or by others. For the same person

(see Figure 22), annotations on the positions of their hands are used to check for occlusions caused by them. In the case of a frontal pose, the positions of both hands are examined, while for the left profile, occlusion by the left hand is evaluated, and for the right profile, by the right hand. Although certain cases are excluded, the DAD-3DNet model is capable of handling some partial occlusions. However, there may be situations where a total loss of visibility cannot be avoided because not all annotations are considered in the process, as shown in Figure 22.

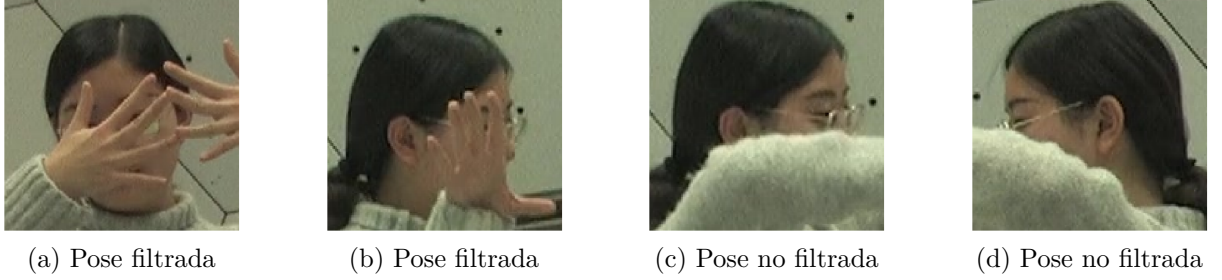


Figure 22: Occlusions between the same person.

Regarding occlusions caused between individuals, it is necessary to check whether the coordinates of the points on the face or hands coincide with the face of another person. If there is an overlap between faces, both will be identified as occlusions in that view to prevent possible errors by the model when interpreting the presence of two faces together. In Figure 23, two examples are shown where the images will be identified as occlusions, and these camera views will not be used in the future reconstruction.



Figure 23: Filtered occlusions between individuals.

If a face is identified as valid and visible, it is then cropped from the original image, taking into account both the initially detected orientation and the positions of the eyes, nose, and lips. Although the size of the extracted images may vary to adapt to the head size in the image, a square format is maintained, as this is the format required by DAD-3DNet and provides better adaptation to different face orientations. In addition to this purpose, this filtering also prevents using cameras that are not functioning, as they will have no associated annotations.

4.2. 3D reconstruction and annotation of the Panoptic database

The following sections detail both the process for performing 3D reconstruction using multiple camera views from the same frame and the algorithms developed for ordering

and selecting these views. The goal is to investigate how the arrangement and selective omission of some cameras affect the quality of the final reconstruction. Information is provided on the methods used for evaluating cameras, the error criteria applied, and, finally, the process for obtaining the annotations for the database.

4.2.1. View predictions with DAD-3DNet

After applying the preprocessing described above for each frame and filtering the camera views, the selected images are used for 3D reconstruction. The first step is to obtain the predictions of the 5023 landmarks for each view using DAD-3DNet. These data are essential for performing the linear initialization and the refinement with BA.

For example, in the case of evaluation scene number 8, out of the 31 cameras, 11 images were selected in which the subject’s face is clearly visible. By applying DAD-3DNet to the three images in Figure 20 corresponding to the same frame, the meshes represented in Figure 24 are generated.

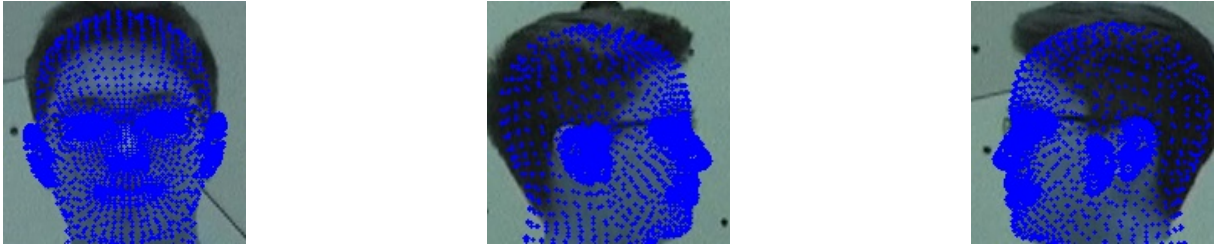


Figure 24: Facial landmark detections with DAD-3DNet.

However, it is important to note that because the data used are predictions from a neural network based on a single image, some of the reference points may contain errors, as seen in the examples in Figure 25 for different scenes.

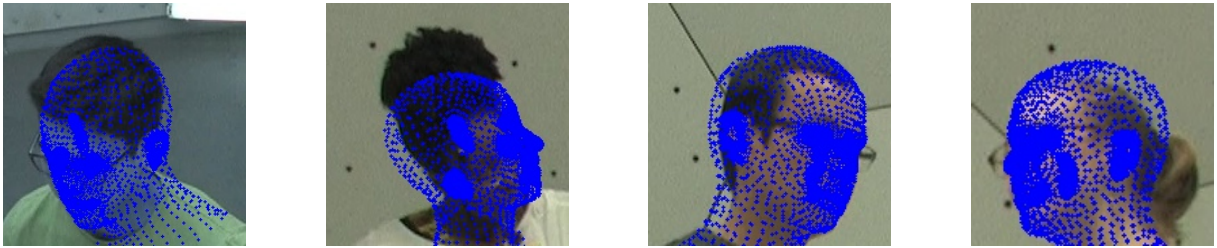


Figure 25: Facial landmarks detected by DAD-3DNet showing considerable error

Since these are the data used for the reconstruction process of a frame, special attention must be paid to their quality in each image to avoid using incorrect predictions in the final reconstruction. This analysis procedure will be examined in detail later.

4.2.2. 3D facial reconstruction

The 3D reconstruction method is based on using the landmarks detected by DAD-3DNet to obtain a refined 3D model. This algorithm uses this set to construct the head model through the method described in section 3.4.1. Then, using the BA algorithm, developed in section 3.4.2, this facial reconstruction is adjusted by applying the camera calibration parameters provided in Panoptic, modifying only the positions of the points in space.

4.2.3. Camera sorting algorithms

In order to obtain higher-quality reconstructions, several methods have been studied to order the predictions associated with each camera according to their accuracy with respect to reference data. As detailed in section 5.1, when all cameras are used in the reconstruction, the error tends to be higher than if a prior selection is performed. To find the optimal arrangement of cameras and avoid incorrect predictions, three algorithms have been developed, which will be compared alongside random camera ordering. These algorithms organize the cameras of the evaluation scenes, and their results will be presented in section 5, justifying the algorithm used for obtaining the final 3D reconstructions.

The following describes the functioning of each of the three proposed methods, detailing their characteristics and operation. For each of them, a linear reconstruction must first be performed as an initialization for BA.

Pairwise camera sorting against Panoptic

In this method, the views (or cameras) are ordered using the reconstruction error (Equation 10) for each pair of cameras. The ordering criterion is the average of the previous errors for each view compared to the others. For each camera, its linear reconstruction is calculated with another camera and refined using BA. After this procedure, its error with respect to the 27 landmarks obtained from Panoptic annotations is recorded, and this error value is saved for the two cameras involved. Finally, the average reconstruction error for each camera is computed, and the views are ordered from lowest to highest according to these values. In this algorithm, a reconstruction is performed for each pair of cameras, so the execution time increases according to $\mathcal{O}(n \cdot (n - 1))$, where n is the total number of views.

Camera sorting using RANSAC against Panoptic

This procedure is the final algorithm selected for camera ordering and selection. It uses the same idea as the previous method, obtaining the error with respect to Panoptic annotations, but applies a random sampling algorithm (RANSAC-style) to reduce execution time. Its operation is as follows:

1. The minimum error for considering two cameras as *inliers* is calculated. To obtain this value, the reconstruction is built and refined using all cameras ordered randomly. As seen in section 5.1, the error value stabilizes when all cameras are used regardless of order, and this value is usually higher than when only two cameras are used.
2. During 50 iterations, two cameras are randomly selected, and their linear reconstruction is obtained and refined with BA. Note that due to the previous camera preprocessing, roughly half of the 31 HD cameras are often removed due to their arrangement in Panoptic (Figure 9) or other occlusions. This number of iterations was compared to using 30 and 40 iterations, showing that the error could increase significantly with more cameras. Using higher numbers of iterations, such as 60, 80, or 100, stabilizes the error, but the execution time is not justified by the slight reduction in error. This is because the best predictions usually produce an error

lower than the minimum error when combined with other cameras thanks to [BA](#) adjustment, so they are still considered *inliers*. However, combinations of cameras with higher mutual errors tend to exceed the minimum error. Although errors may occur due to the algorithm’s random nature, this number of iterations is considered sufficient for the analysis performed.

3. If the reconstruction error relative to the selected Panoptic landmarks is lower than the minimum error, the cameras are considered *inliers* for that iteration. Their individual counters are incremented to indicate how many times they have met this condition.

After completing the iterations, the number of times each camera has been an *inlier* is obtained, allowing the views to be ordered from highest to lowest according to this value.

Camera sorting using [RANSAC](#) against DAD-3DNet predictions

In this case, the procedure is the same as above, but the error is computed using the 5023 landmarks from the DAD-3DNet predictions. This comparison has the advantage of considering all points of the head, not just a few from the frontal face. However, regarding the application of the methods, it is important to remember that both DAD-3DNet predictions and Panoptic annotations may contain errors in the key points, so these criteria can lead to inaccurate results since they are not manually annotated.

4.2.4. Camera selection for the final 3D facial reconstruction

After evaluating the aforementioned methods, whose analysis is presented in section 5, the [RANSAC](#) method against Panoptic annotations was selected. This choice is based on its generally lower errors and reduced execution time compared to checking all views with more than seven cameras. Compared to the method using DAD-3DNet predictions and random camera ordering, this error difference is even more pronounced in most cases.

Up to this point, these methods were limited to ordering cameras or, more precisely, predictions from each camera. However, it is also necessary to avoid using certain views that could harm the final model. For this camera selection, the list of how many times each camera provided a low-error reconstruction is used according to the following criteria:

- If no camera has been an *inlier*, i.e., no pair of cameras achieved a lower error than using all cameras, all views are used in the reconstruction in their initial order.
- If only 2 or 3 cameras have been *inliers* (note that a single camera is never sufficient), only these predictions are used in the reconstruction.
- If there are 4 or more cameras that have been *inliers*, the number of these cameras that were *inliers* more than 3 times is counted. If at least 4 cameras meet this criterion, only the predictions from these views are selected for the final reconstruction.
- If fewer than 4 cameras from the selected set have been *inliers* 3 times or more, the reconstruction is performed using all cameras that were *inliers* at least once.

The goal of this selection is to obtain the set of cameras that have been *inliers* multiple times during the iterations. However, if very few cameras are considered *inliers*, or if they have been identified only a few times, all cameras that have ever been *inliers* are used. This occurs when a set of cameras has considerably higher errors, preventing their pairing with better views from producing a lower combined error. In this way, these cameras are considered *inliers* only when paired among themselves. Additionally, a stricter filter is applied to select fewer cameras when some consistently achieve significantly lower errors in the reconstructions they generate, based on the number of times they were *inliers* with other cameras.

4.2.5. Facial annotations of the Panoptic dataset

The XYZ points in 3D space that form the final refined reconstructions are saved in TXT format, making them available for training new models. Similarly, their reprojection onto the corresponding images is saved in normalized JSON format, producing labels corresponding to the 2D facial landmarks. This normalization is performed between 0 and 1 and allows the landmarks to be adjusted when the image size changes. This processing is necessary because not all cropped images have the same size and must be resized for the training of landmark detectors, allowing the points to be aligned with the new dimensions. For the reprojections used in sections 4.3 and 4.4, only a portion of these annotations is selected. Nevertheless, it should be noted that although the data has been processed and refined, it is not as accurate as if manually annotated or obtained using laser scanners.

4.3. Extracting the visibility of annotated 2D landmarks

The 3D head reconstructions are intended to be used for training facial analysis models. In this project, these annotations will be validated using two landmark detectors, one of which will consider the visibility of the 2D landmarks during training. This concept of landmark visibility depends on the head pose of each subject. For example, in a frontal image, occlusions will occur at the back and top of the head. However, in a profile image, these occlusions will correspond to the opposite side of the head shown in the image. Therefore, to extract these points, the following procedure is followed:

- item The *yaw*, *pitch*, and *roll* angles corresponding to the head orientation provided by the DAD-3DNet model are extracted, and the corresponding rotation matrix is calculated. Figure 26 shows the detected orientation of the subject in the image.



Figure 26: Subject orientation in the image.

1. The midpoint of the 5023 points of the reconstruction is obtained, represented as m in Figure 27.

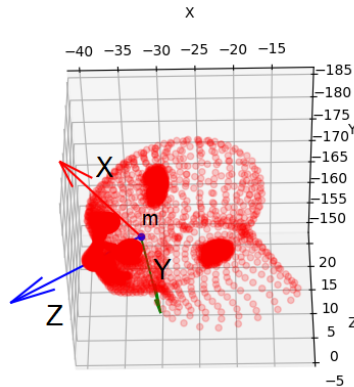


Figure 27: Orientation vectors on the reconstruction.

2. The rotation matrix R is calculated between the reference system of the 3D head model and that of the camera. The axes of the head reference system, which are the columns of R , are shown from the midpoint of the reconstruction in this model. In Figure 27, these detected orientation axes referenced to the 3D model are visible. Blue represents the Z axis, green the Y axis, and red the X axis, indicating the direction the subject is facing.
3. To construct the 3D reference system in space, the vector H defining the 3D head axis is first obtained. Using the midpoint and selecting landmark number 3565, this vector is defined, shown in black in Figure 28, providing information about the head's tilt in space.

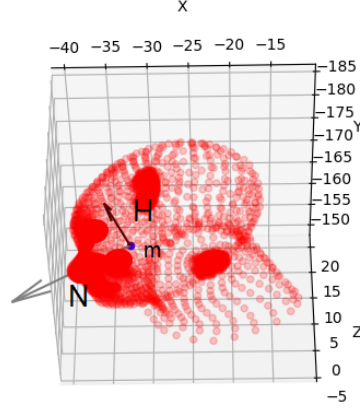


Figure 28: 3D reference system.

4. Next, the axis N indicating the orientation of the 3D model is obtained, which corresponds to the X axis of the reconstruction. For this, point number 3526, located on the nose, is selected, and the vector from the midpoint to the nose is calculated, constituting vector N . This defines the 3D reference system in space as shown in Figure 28.
5. The angular difference in degrees between the image X axis and the corresponding N axis is calculated.
6. The vector P , which lies on the plane dividing landmark visibility, is obtained. To compute this vector, the X vector is rotated around the H axis by twice the previously calculated difference. This rotation is clockwise or counterclockwise depending on whether the X axis is on the right or left side of the 3D model. In this way, an analogous vector toward the opposite side is obtained. Then, an additional 90° is added or subtracted, yielding vector P , which is part of the plane dividing points into visible or non-visible. This is shown in Figure 29 in magenta.

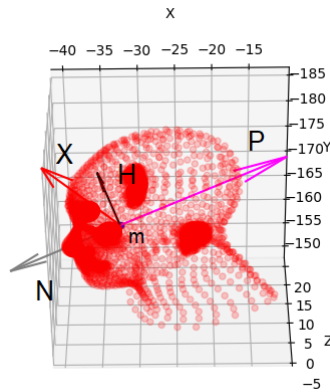


Figure 29: Vector contained in the plane.

7. The plane is constructed using the H axis of the 3D reconstruction and the plane vector P . This plane is shown in Figure 30.

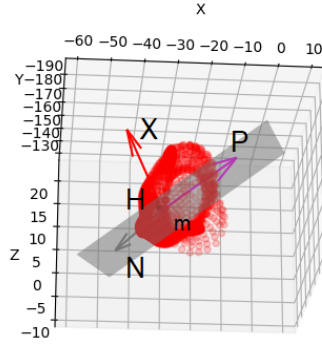


Figure 30: Landmark visibility plane.

8. For each point, it is determined on which side of the plane it lies, defining its visibility.

After completing these steps, the visibility or occlusion of each point can be extracted according to the previously defined plane. In Figure 31, visible points are shown in red and occluded points in black for each image, along with the orientation vector X , the plane vector P , and axes N and H .

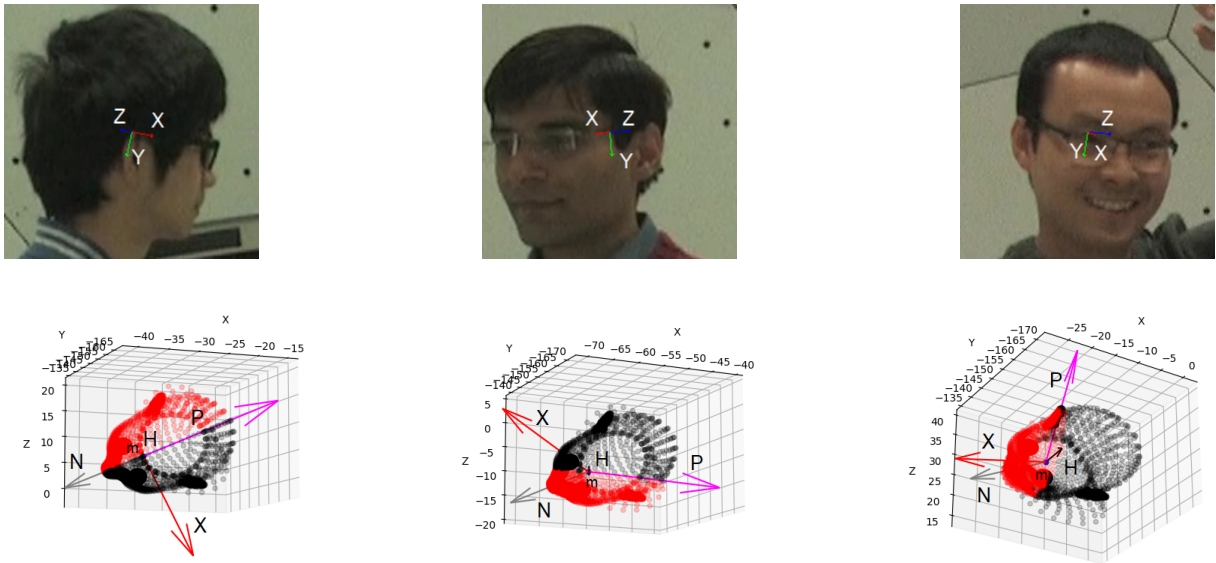


Figure 31: Extraction of visible points according to head orientation.

It is important to note that the division is always performed with respect to the plane along the central axis, without taking into account whether the subject's head is tilted. Consequently, there may be errors in classifying certain points as non-visible in some cases. Additionally, this calculation depends on the accuracy of the rotation matrix predicted by DAD-3DNet.

4.4. Training of 2D landmark detectors

Once the classification of head landmarks into visible and occluded has been obtained, the training of a new model defined in Keras begins. The input to this model is a cropped

image of a subject's head with a size of 256×256 , while the expected output corresponds to the 5023 characteristic points defining the shape of the head, as obtained with DAD-3DNet.

As described in subsection 3.7.2, two training approaches are carried out: one where the loss is equal for all points, and another where the loss of occluded points is set to 30 % of the point difference (Equation 15), reducing their impact on the final prediction.

For this part, a total of 1568 images from the scenes *170404haggling_{b1}*,

However, it is important to remember that, due to the nature of the labels used, as they are automatically processed annotations, they may contain errors. Figure 32 shows some examples of these cases, where the errors may be caused by a poorly processed reconstruction or an incorrect orientation obtained by DAD-3DNet. These errors will clearly affect the performance of the models, influencing the loss function calculated according to the visibility of the points.



Figure 32: Example of incorrect images used in training with their labels.

On the other hand, to perform training with the custom loss considering the hidden landmarks in the image and the batch size, the following procedure is followed:

1. Four variables are declared using the Keras backend to store the indices of the visible and hidden landmarks for training.
2. One of the variables stores the list of indices of the visible landmarks for the current batch, while another stores the list of hidden indices. Regarding the other two variables, only the visible and hidden indices corresponding to the last batch are stored. This step is necessary because the last batch may have a different size from the previous ones, and it is important to handle these size changes during training with TensorFlow.
3. Using a training callback function, the indices of the visible and hidden landmarks in the Keras variables are updated at the end of each batch with the values for the new set of images.

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Chapter 5

Experimental results

This chapter presents the results obtained during the evaluation of the camera selection and sorting algorithms in the final 3D reconstructions. The results of the training of the landmark detectors using the new annotations are also analyzed.

For the first part, this section is divided into a general evaluation of the methods and a specific evaluation of the selected algorithm. In this general reconstruction evaluation, the variation of the error is analyzed with respect to the group of cameras used in this process, both against the annotations provided by Panoptic and the manually created annotations. It is important to highlight that this evaluation is carried out on the evaluation scenes against the CMU Panoptic annotations and the manually created annotations. Although all the annotations from the Panoptic database are available, the manual evaluation is performed only on a set of 15 selected scenes (see Figure 17) with their respective views, covering a total of 183 images. These will be referenced as presented in Table 1.

Once the best camera sorting algorithm is chosen, a specific evaluation is presented on the final refined reconstructions through a more detailed analysis of the evaluation scenes. Finally, these results are shown through the texturing of the 3D models.

In the section on landmark detector training, graphs of the loss function values for the new models are presented and compared with the predictions of the DAD-3DNet neural network and the final annotations used.

5.1. Evaluation of view sorting methods

This general evaluation presents the results of the study carried out for selecting the developed camera sorting algorithm. As detailed in Section 4.2.3, three methods were executed and evaluated in comparison with the use of a random camera ordering to determine the best reconstruction strategy. First, one of the key factors of these algorithms is the execution time associated with each for obtaining the prediction ordering. Table 2 shows the average times in seconds to obtain the camera order in each case.

Table 2: Average time to sort the cameras by each method.

	Average time (s)
Sorting with pairwise comparison and Panoptic	88.76
Sorting with RANSAC and Panoptic	30.36
Sorting with RANSAC and predictions	42.17
Random sorting	2.61e-05

Since the random sorting algorithm uses only a single function, its execution is immediate. A difference of more than 40 seconds is also observed for the [RANSAC](#) methods compared to checking all cameras in pairs. In this case, execution takes about a minute and a half to complete the camera (or view) ordering.

In addition to this study, the errors obtained after performing several reconstructions with each method will be analyzed in comparison with the Panoptic annotations and the manually annotated evaluation scenes. Therefore, it is important to emphasize that the evaluation presented in this section does not correspond to that of the selected final reconstruction. The purpose of this section is to objectively show why both the ordering and the choice of the number of cameras are key parameters in selecting the final reconstruction.

To carry out this process, the proposed algorithms are first used to sort the cameras according to each algorithm’s criterion. Subsequently, for each ordered set of cameras, refinement is performed using [BA](#), adding the remaining cameras in order and obtaining the corresponding error for each case. That is, for a linear initialization with two cameras, refinement is carried out with 2, 3, 4... cameras; then, using three cameras in the initial linear reconstruction, refinement is similarly performed starting with 2, 3, 4... cameras up to the total. This procedure is applied to all available camera views of the same frame.

In this way, the influence of the ordering of the best predictions on the 3D model is studied, as well as how variation in the total number of views used in initialization or refinement can affect the quality of the final results.

5.1.1. Evaluation of methods against Panoptic annotations

To evaluate the reconstructions in comparison with the Panoptic annotations, it is necessary to compare the positions of the 27 landmarks originally annotated in Panoptic (see [Figure 12](#)) with the corresponding reprojected points from a reconstruction. The error is determined as the average of all [RMSE](#) reprojection errors obtained for the 15 evaluation scenes after reconstructions performed with different numbers of cameras in initialization and refinement, as explained above. Therefore, the position difference between the annotated landmarks and the refined landmarks is compared, and the reprojection error in pixels is calculated for each method according to [Equation 12](#). [Table 3](#) presents this error obtained from all reconstruction combinations for all evaluated scenes in each algorithm.

Table 3: Reconstructions error for each method compared to Panoptic.

	$\mathcal{E}_m$ (píxel)
Sorted reconstructions with camera pairs and Panpotic	37.16
Sorted reconstructions with RANSAC and Panoptic	26.65
Sorted reconstructions with RANSAC and predictions	56.02
Randomly sorted reconstructions	51.55

In general, a lower error is observed when cameras are ordered based on their error relative to Panoptic. This makes sense since the current evaluation is effectively being performed against the Panoptic annotations. It is observed that using a random order can even be beneficial compared to using comparisons with DAD-3DNet predictions. The lowest error is obtained with the [RANSAC](#) method using Panoptic, which also has a shorter execution time.

To obtain the error calculated for each individual scene, Equation 11 is used. In the graph in Figure 33, the $\mathcal{E}_e$ for each scene, e , are shown.

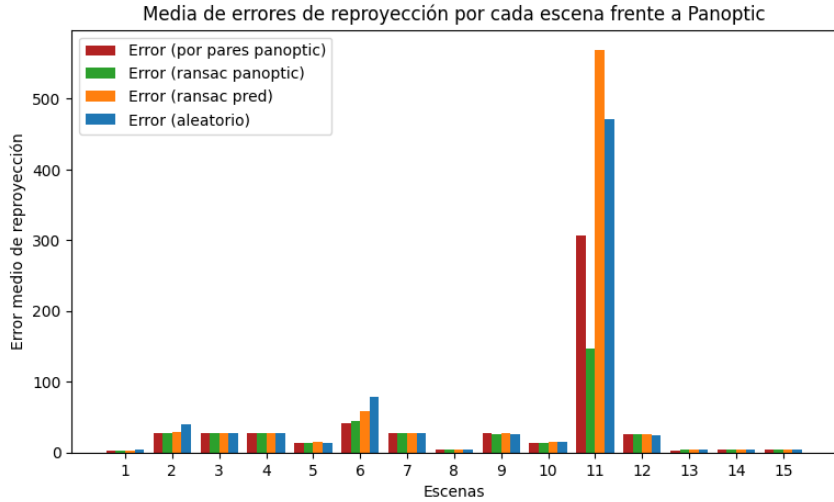


Figure 33: Reprojection error for each scene against Panoptic.

A similar trend can be seen across the scenes, where both the random method and the prediction-based method tend to have higher errors. It is worth noting, however, that the error tends to stabilize when averaging over many combinations. A noticeable difference appears for scene 2, thanks to avoiding the use of a camera that produces a large error in most reconstructions. However, this camera is used with the random sorting method. It should be remembered that the refinement of points depends on the DAD-3DNet predictions themselves, so when these predictions have larger errors, the refinement will also reflect them, as is the case for scene 11. Furthermore, since these evaluations are performed only on 27 landmarks, the full influence of camera ordering cannot be completely observed.

To understand exactly where these average errors come from, scene *171026_pose1*, *corresponding to identity*

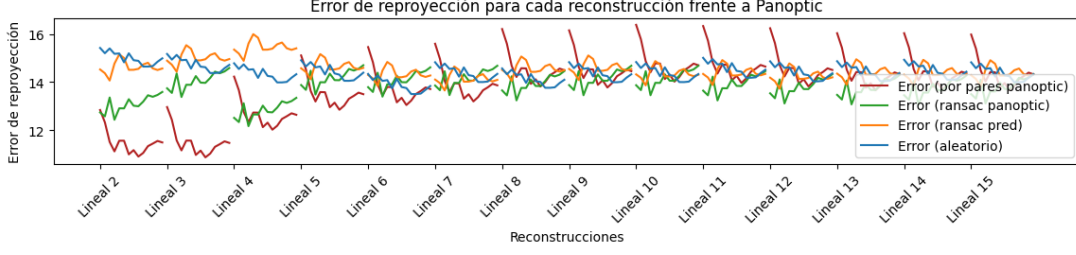


Figure 34: Reprojection error for each reconstruction against Panoptic (scene 10).

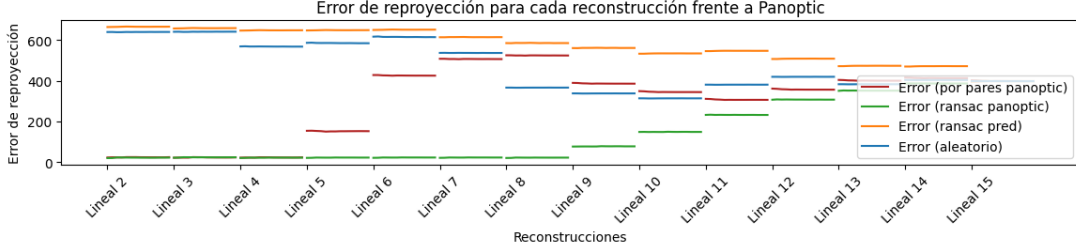


Figure 35: Reprojection error for each reconstruction against Panoptic (scene 11).

It is clearly observed that the lowest error is achieved in most cases when cameras are ordered according to their error compared with Panoptic, while the highest errors are found in the random sorting algorithm and the comparison method with DAD-3DNet. These differences can be more than 50 pixels, as in Figure ??, just a few pixels, as in Figure 34, or even hundreds of pixels (Figure 35). In the case of scene 2, this error is influenced by the use of camera number 13 in the reconstructions, whose prediction error is higher. This camera is not filtered by the random method and is used from the beginning in the reconstructions. In the case of scenes 10 and 11, the pairwise method has lower errors initially. However, after adding five cameras, a prediction with larger errors is included. This situation is avoided using the minimum error and the **RANSAC** iterations.

It should be noted that both the order and the number of cameras used in each case indeed have an influence, with lower errors obtained when fewer cameras are used in a specific order. In addition, the reconstruction error stabilizes when more cameras are used regardless of the order, although the errors are higher than when selecting fewer cameras. This is why this error value is used in the **RANSAC** algorithms as the minimum error to consider a camera *inlier* in the iteration.

Similarly, this comparison is made against the annotations provided by Panoptic, which may be incorrect in some cases, leading to situations where worse-positioned points have a lower error in comparison.

5.1.2. Evaluation of methods against manual annotations

In order to obtain a more accurate and reliable evaluation, the same approach as above is used, but the error is calculated against the manually created annotations. In Table 4, the mean error calculated with Equation 12 ($\mathcal{E}_m$) for all reconstructions across all methods is presented.

Table 4: Error of reconstructions for each method versus manual annotation.

	$\mathcal{E}_m$ (píxel)
Sorted reconstructions with camera pairs and Panoptic	24.82
Sorted reconstructions with RANSAC and Panoptic	14.77
Sorted reconstructions with RANSAC and predictions	43.62
Randomly sorted reconstructions	39.17

The most notable aspect of Table 4 compared with Table 3 is that the mean error of the sets is much lower, which is beneficial since it corresponds to the error against the real ground-truth. Likewise, lower errors are achieved with the methods that compare against Panoptic than when using predictions or a random camera order, indicating that the trend is maintained and there is an advantage in ordering cameras using these annotations. If we break down the mean error by scenes as in Figure 36, the different values between frames can be observed.

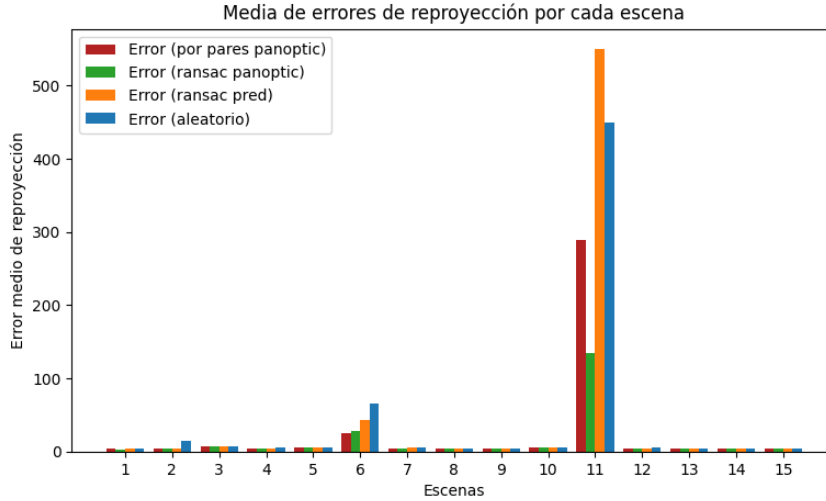


Figure 36: Reprojection error for each scene against manual annotations.

In most cases, the errors of the methods remain consistent across the scenes, although some differences exist, with lower errors still achieved by the methods compared to Panoptic. The error in scene 2 also shows the characteristic previously mentioned. For scenes 6 and 11, higher errors are obtained because the predictions provided by DAD-3DNet contain larger errors relative to the manual ground-truth than other scenes. It is important to note that, in some cases, using the pairwise comparison methods can result in biased data, since only the error provided by two cameras is considered to obtain a reconstruction, so some views with worse predictions can achieve lower errors in combination and be positioned earlier in the sorting list. Likewise, Panoptic annotations can also contain errors and significantly affect these results.

To better understand the reprojection error of each reconstruction for each method, the errors $\mathcal{E}_r$ for scenes 2, 4, and 6 are broken down. Following the same color legend, some conclusions can be drawn from Figures 37, 38, and 39. In Figure 37, it is clearly seen that the addition of camera 13 causes the error to increase considerably in the reconstruction. Using the sorting algorithms ensures that this camera is positioned at the end, which is when this increase in error is observed. However, when using a random method, the

predictions provided by camera 13 are positioned at the beginning. In the rest of the graphs, the trend is the same, with the random sorting error generally being higher, since cameras with worse predictions are not placed at the end of the list, whereas the proposed methods achieve lower errors with fewer cameras.

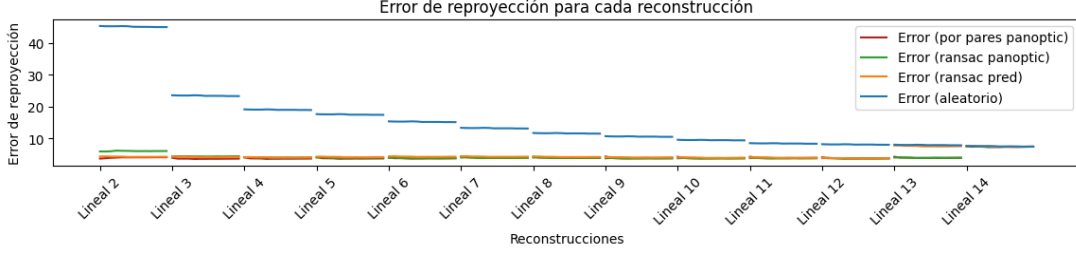


Figure 37: Reprojection error for each reconstruction versus manual annotations (scene 2).

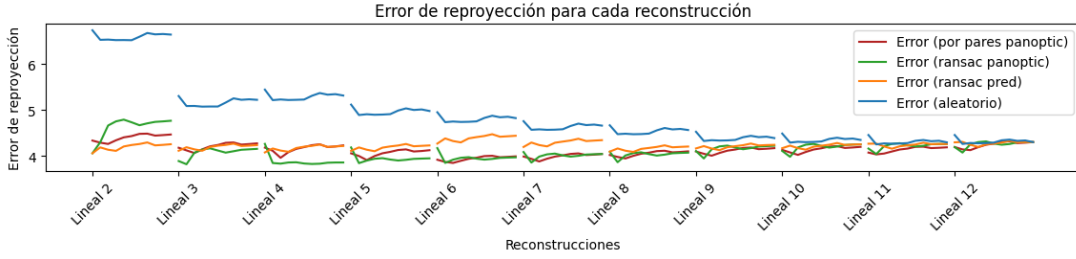


Figure 38: Reprojection error for each reconstruction versus manual annotations (scene 4).

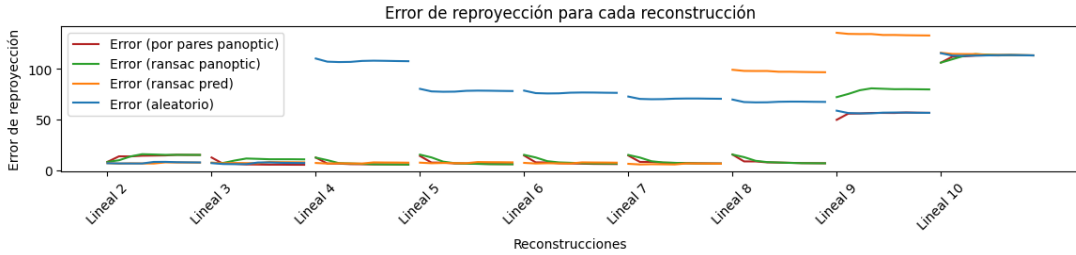


Figure 39: Reprojection error for each reconstruction versus manual annotations (scene 6).

Therefore, it is highlighted that the best reconstructions are obtained with approximately 4 to 7 cameras, where the smallest errors are observed for all methods. The error progressively increases when more images are used in the initial reconstruction. Similarly, there is a difference when using a different number of cameras in the BA refinement with the same initialization, although the trend remains consistent across all algorithms, as observed in the previous comparisons.

In conclusion, the method using RANSAC evaluated with Panoptic is selected for reconstruction because it achieves a lower overall error. Although it may produce slightly higher errors in some cases, the difference is minimal—only a few pixels compared to the other methods, as shown in the previous graphs. This potential effect is not noticeable in the final reconstructions, and the execution time is considerably shorter than using pairwise comparison when the current frame contains more than 7 cameras.

It should be noted that the conclusions drawn throughout this analysis served as the basis for selecting or omitting cameras according to the *inliers* list provided by [RANSAC](#). Specifically, this study guided the procedure explained in Section 4.2.4, which must select at least 4 cameras for the final reconstruction. In this way, the number of cameras when the error is lower and the *inlier* cameras associated with these reconstructions using the developed [RANSAC](#) with Panoptic algorithm were taken into account.

5.2. Evaluation of the final reconstruction

After deciding to use the [RANSAC](#) method in combination with Panoptic annotations according to the previous evaluation, the procedure explained in Section 4.2.4 is followed, generating a single reconstruction for each frame of a scene. This reconstruction uses the same number of cameras and in the same order for both the initialization with [DLT](#) and the refinement with [BA](#).

In this section, the quality of the 3D model is analyzed in detail in comparison with using all views ordered randomly, providing an evaluation similar to that carried out in the previous section.

5.2.1. Evaluation of the final reconstruction against Panoptic

First, in Figure 40, the errors of each scene ($\mathcal{E}_e$) of the final reconstruction are shown compared with the random reconstruction according to the error with Panoptic annotations.

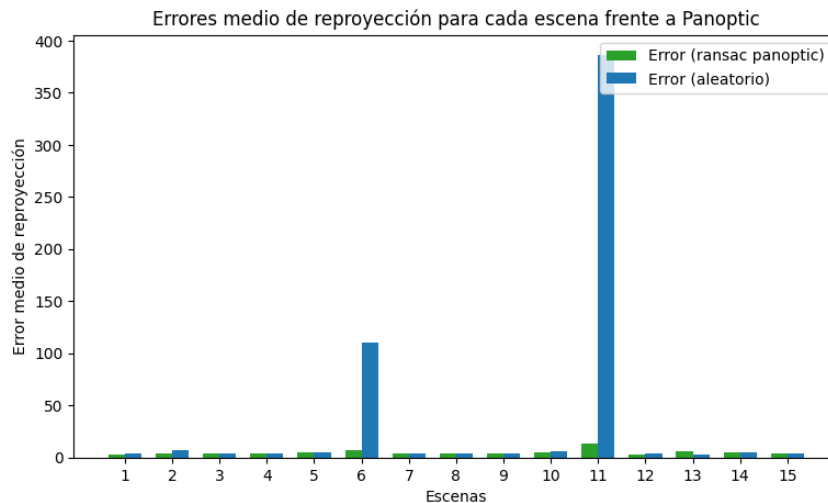


Figure 40: Reprojection error of the final reconstruction for each scene in front of Panoptic.

It can be seen that, in general, the error tends to be higher when using a random camera order, although the difference is usually not large. For scenes 6 and 11, the pixel error is also significantly higher compared to other scenes. This occurs because the DAD-3DNet predictions contain larger errors for some of them. Consequently, the error for facial points is reduced with the sorting method, although the reconstruction is also affected.

This reprojection error can be analyzed for each view of a specific scene using Equation 9. Scenes 11 and 12 are selected, where the different values between images can be clearly observed. These errors ($\mathcal{E}_c$) are shown for each camera in Figures 41 and 42.

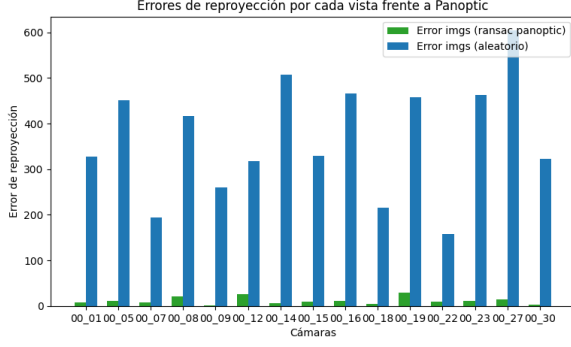


Figure 41: Reprojection error of the final reconstruction for each camera in front of Panoptic (scene 11).

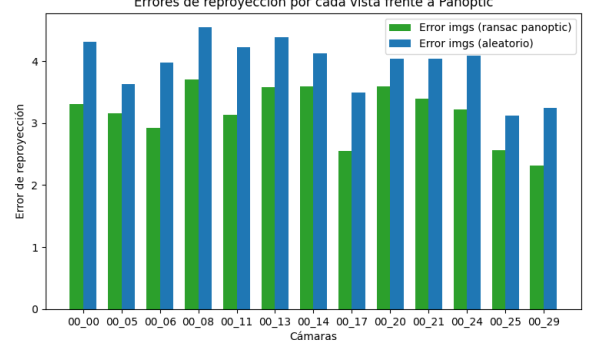


Figure 42: Reprojection error of the final reconstruction for each camera in front of Panoptic (scene 12).

A significant difference between the two methods can be observed in scene 11. For scene 12, a lower error rate is also maintained with **RANSAC** and Panoptic, but the difference is only about 2 pixels at most, so it is not a significant error. Continuing with the analysis of scene 11, the error per landmark can be studied using Equation 7. This error ($\mathcal{E}_{le}$) for the key points of scene 11 is presented in Figure 43 to understand which points show the greatest detection differences.

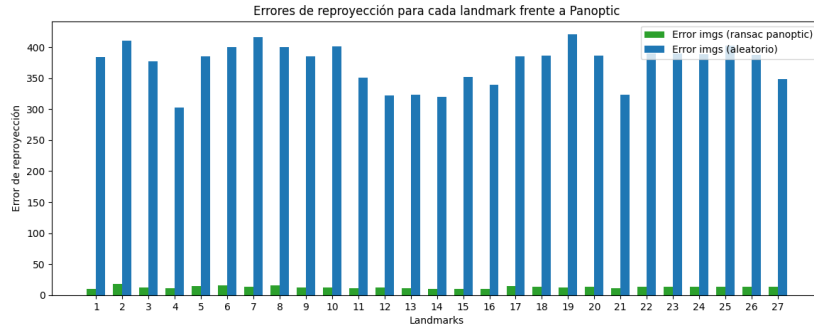


Figure 43: Reprojection error for each landmark against Panoptic (scene 11).

For example, in this specific scene, the key points with the highest error are 2, 7, and 19, corresponding to an eyebrow point, the right eye, and the nose (see Figure 18).

To begin the comparative evaluation against the manually created annotations, the error calculated for each scene using Equation 11 is shown, using **RANSAC** and Panoptic as well as random sorting (Figure 44). This error is also lower compared to the Panoptic annotations shown in Figure 40. Furthermore, a similar pattern in the trend of results across all scenes is observed.

5.2.2. Evaluation of the final reconstruction against manual annotations



Figure 44: Reprojection error of the final reconstruction for each scene against manual annotations.

In Figure 44, the largest discrepancies in error are observed in scenes 6 and 11, as previously mentioned.

In this section, scenes 2 and 5 are analyzed to more clearly observe the errors when the differences are not as large. In Figures 45 and 46, the reprojection errors for these scenes are shown for each view.

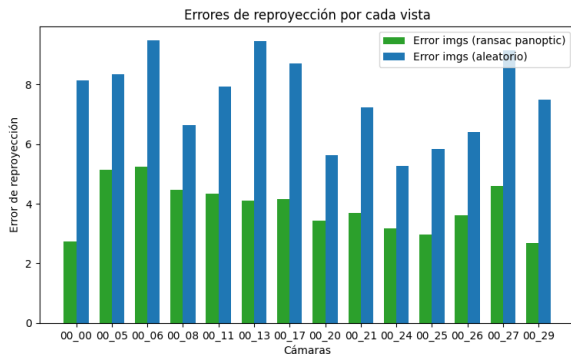


Figure 45: Reprojection error of the final reconstruction for each camera against manual annotations (scene 2).

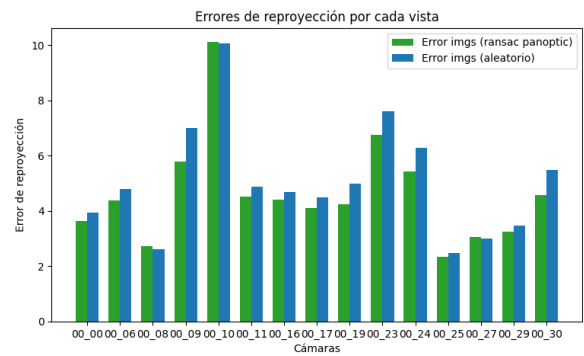


Figure 46: Reprojection error of the final reconstruction for each camera versus manual annotations (scene 5).

For scene 2, larger differences in errors are observed across all views compared to scene 5, although the error is still lower for RANSAC in almost all cases. If the error for each key point is shown for both scenes, the graphs in Figures 47 and 48 are obtained.

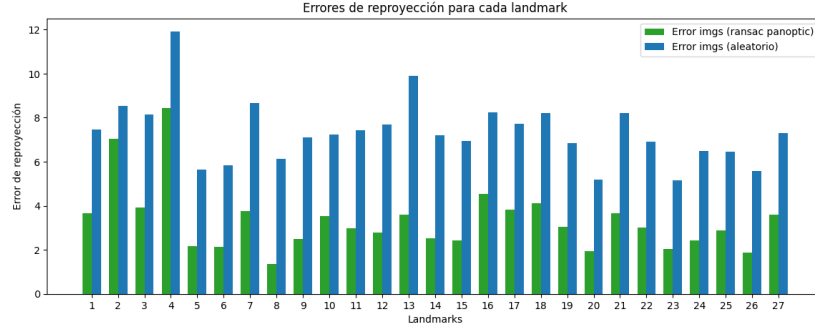


Figure 47: Reprojection error of the final reconstruction for each landmark against manual annotations (scene 2).

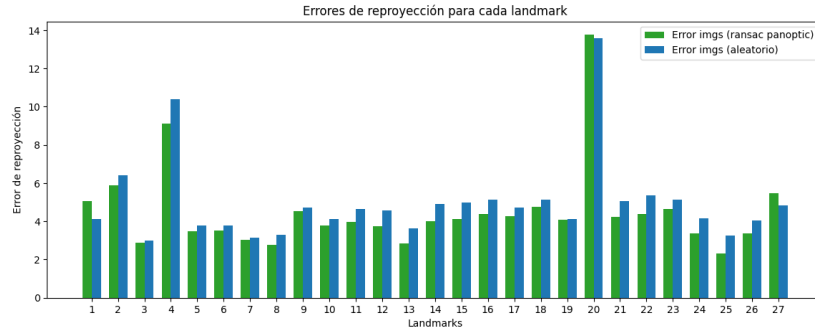


Figure 48: Reprojection error of the final reconstruction for each landmark against manual annotations (scene 5).

Examining the results for the landmarks, it can be concluded that key point number 4, corresponding to the left eyebrow (Figure 18), shows a large error in both cases. In scene 5, a large error is also observed for point 20, located on the nose.

Next, in Figures 49 and 50, several perspectives of scenes 2 and 5 are shown, providing a clearer visualization of these discrepancies. The green points correspond to the selected **RANSAC** algorithm, the blue points to the random sorting, and the white points to the manual ground-truth.

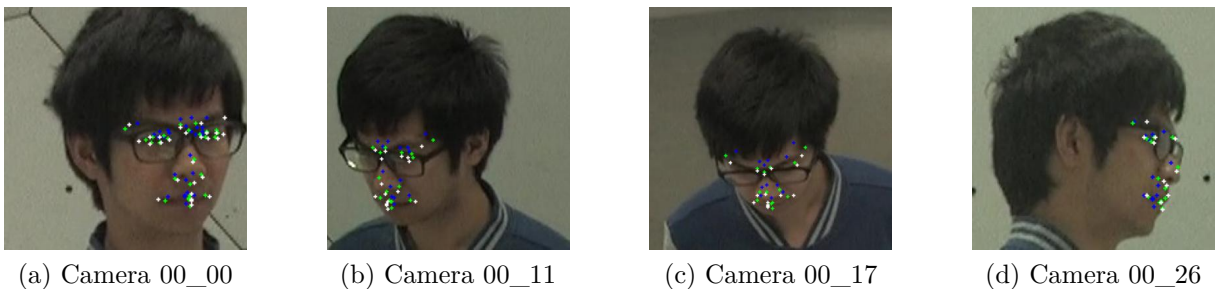


Figure 49: Landmarks comparison (scene 2).

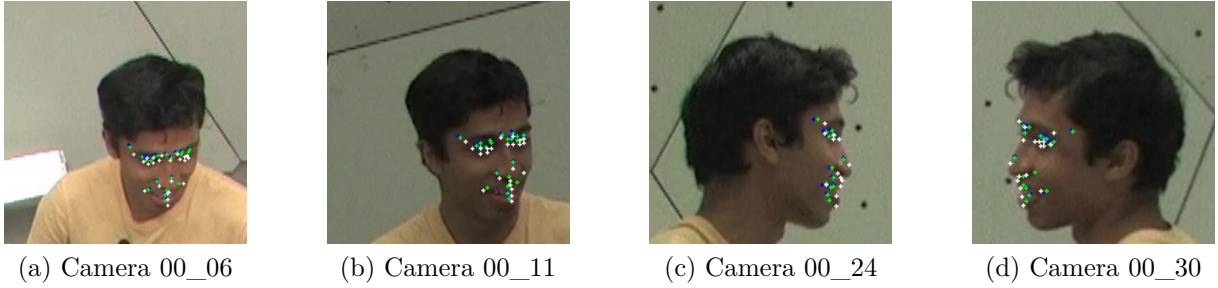


Figure 50: Landmarks comparison (scene 5).

In Figure 49, large differences in the positions of the landmarks can be seen, highlighting the importance of selecting and ordering the cameras. On the other hand, in Figure 50, there is a smaller difference between the green and blue points. Similarly, when errors exist in the landmarks of these images, both sets tend to deviate in a similar manner, although those obtained by RANSAC are generally closer to the ground-truth points, resulting in smaller errors.

5.2.3. Visual evaluation against DAD-3DNet predictions

In previous sections, the impact of camera selection was studied by evaluating the results against a set of 27 facial landmarks. However, the final predictions actually provide a total of 5,023 points, leaving many landmarks that cannot be properly evaluated. Considering this, the goal is to show the reprojections of all points forming a mesh to compare its shape with the original predictions generated by the DAD-3DNet model.

Continuing with scene 2, where a difference in error was observed, the meshes formed by the refined points are presented in comparison with the original results provided by DAD-3DNet in Figure 51. In these images, camera 13 is also shown in the third column, as it was the one that significantly increased the reconstruction error. Its incorrect prediction by DAD-3DNet is highlighted, as well as how this issue is resolved by the processed multiview reconstruction.



Figure 51: The first row shows the refined reprojections using the RANSAC algorithm and Panoptic (scene 2). The second row shows the reprojections predicted by DAD-3DNet (scene 2).

In Figure 52, the 3D meshes of scene 12 are compared. An improvement in the head shape alignment can be seen for all the selected images of this scene.

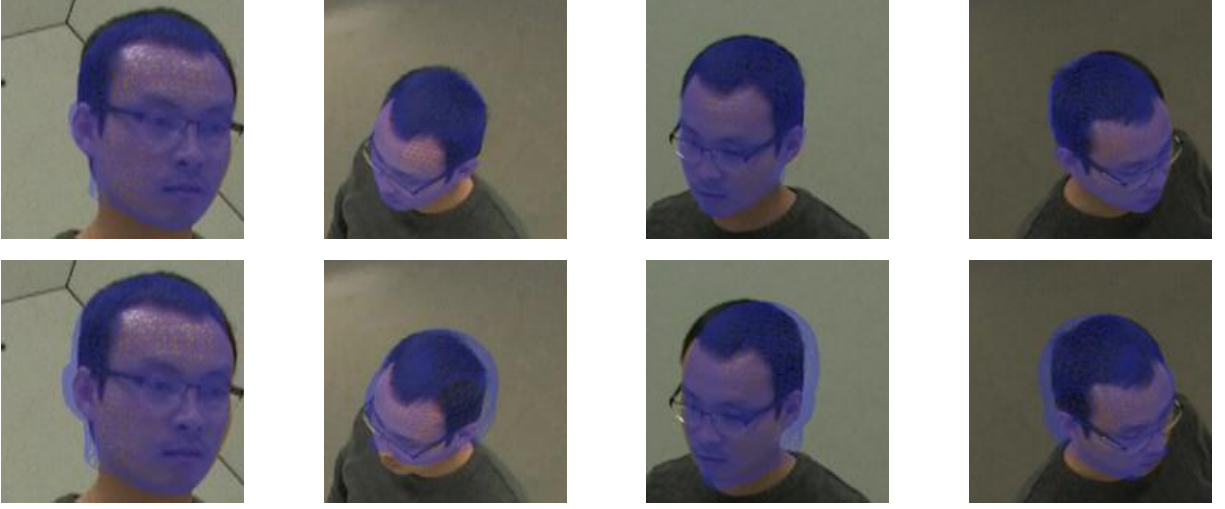


Figure 52: The first row shows the refined reprojections using the RANSAC algorithm and Panoptic (scene 12). The second row shows the reprojections predicted by DAD-3DNet (scene 12).

Among all the figures presented, it should be noted that the reprojections of the refined model come from a single 3D model, while the mesh predicted by DAD-3DNet is calculated independently for each view. Therefore, the [RANSAC](#) refinement allows a reconstruction using the information from all views simultaneously. An improvement is observed when considering predictions from multiple cameras, especially resolving some deviations in the peripheral parts of the head.

However, if the initial DAD-3DNet predictions deviate too much from the true positions of the landmarks, the reconstruction obtained by the proposed method may negatively affect some views. This is the case for scene 11, shown in Figure 53, which already exhibited a large error in the original predictions.

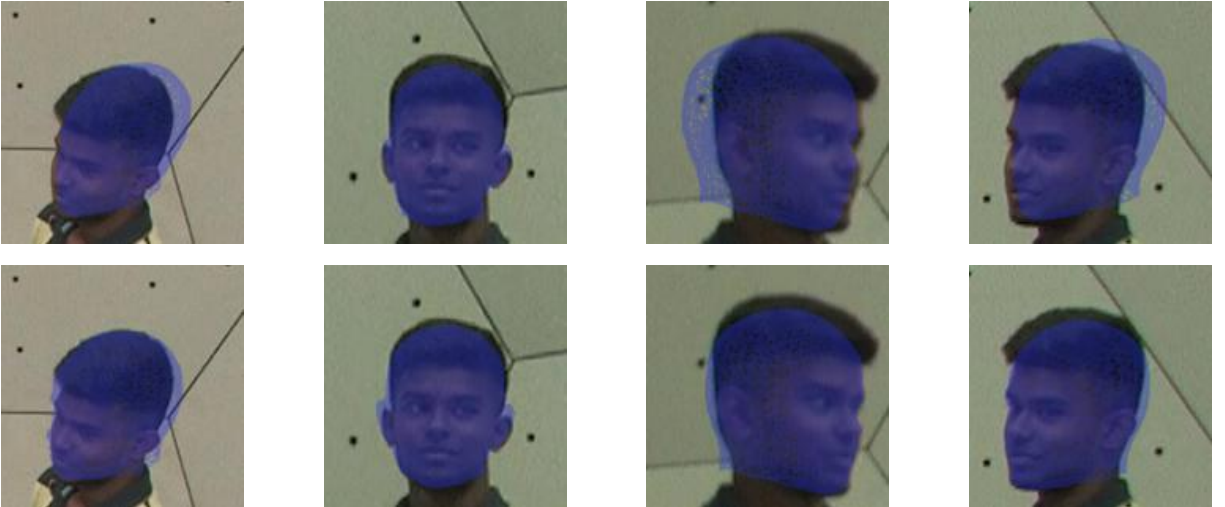


Figure 53: The first row shows the refined reprojections using the RANSAC algorithm and Panoptic (scene 11). The second row shows the reprojections predicted by DAD-3DNet (scene 11).

In the previous scene 11, the facial landmarks were better positioned with the sorting method, but a worse overall mesh arrangement is observed. In the case of scene 6, some of the predictions are adjusted, but small deviations may appear that were not present in the individual image predictions (see Figure 54).



Figure 54: The first row shows the refined reprojections using the RANSAC algorithm and Panoptic (scene 6). The second row shows the reprojections predicted by DAD-3DNet (scene 6).

This can also occur due to deviations in the annotations provided by CMU Panoptic (see Figure 13), since the sorting method uses these points to select the cameras for the final reconstruction.

This section highlights the importance of checking the results with all detections, as the ordering and selection of these predictions is a key factor when performing reconstructions. However, because a limited number of Panoptic landmarks are used and these can be incorrect in some cases, it is likely that some erroneous predictions may be included in the reconstruction. Despite this, if the majority of the selected views have accurate predictions, this issue can be mitigated in images where detections are of lower quality due to complex poses of the person.

5.2.4. Visualization of the final 3D reconstruction

To visualize the complete facial model, the VTK library is used, allowing the quality of the points in 3D space to be inspected. In Figure 55, several examples of images of subject 0 in scene 171204_pose5 are shown.

It can be observed that the 3D model faithfully follows the facial expression, clearly visible in the movement of the lips. However, in some frames with larger errors, there may be deviations in the shape of the head, even though the facial expression is captured correctly. This effect can be seen, for example, in the models shown in Figure 56, which also correspond to the subject from scene 1 for frames similar to the previous ones.

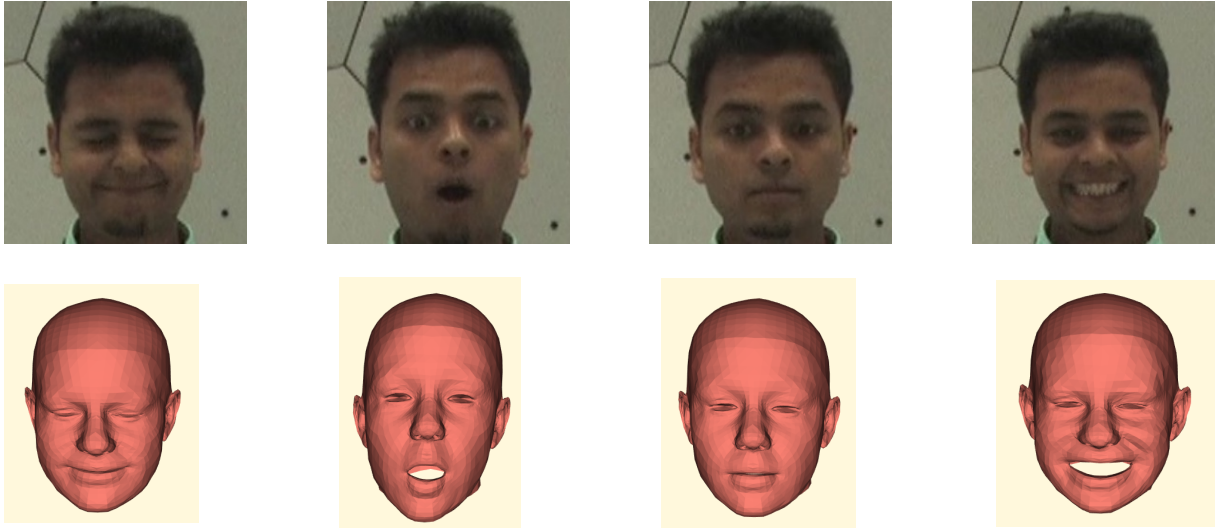


Figure 55: Textured visualization (scene 1).

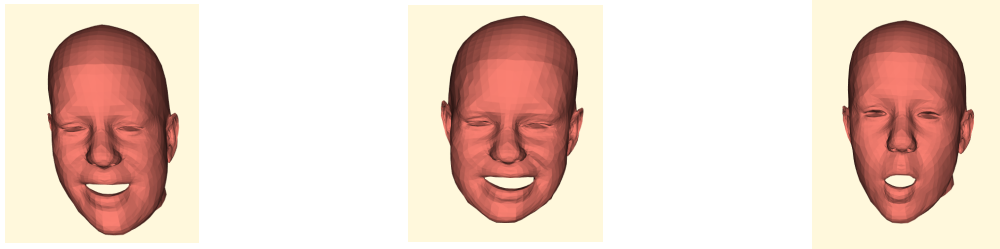


Figure 56: 3D reconstructions failed based on textured model (scene 1).

By applying this texturing to some images from evaluation scenes 3, 9, 10, and 15, Figure 57 shows how the reconstruction conforms to the characteristic shape of each face and its facial expression.

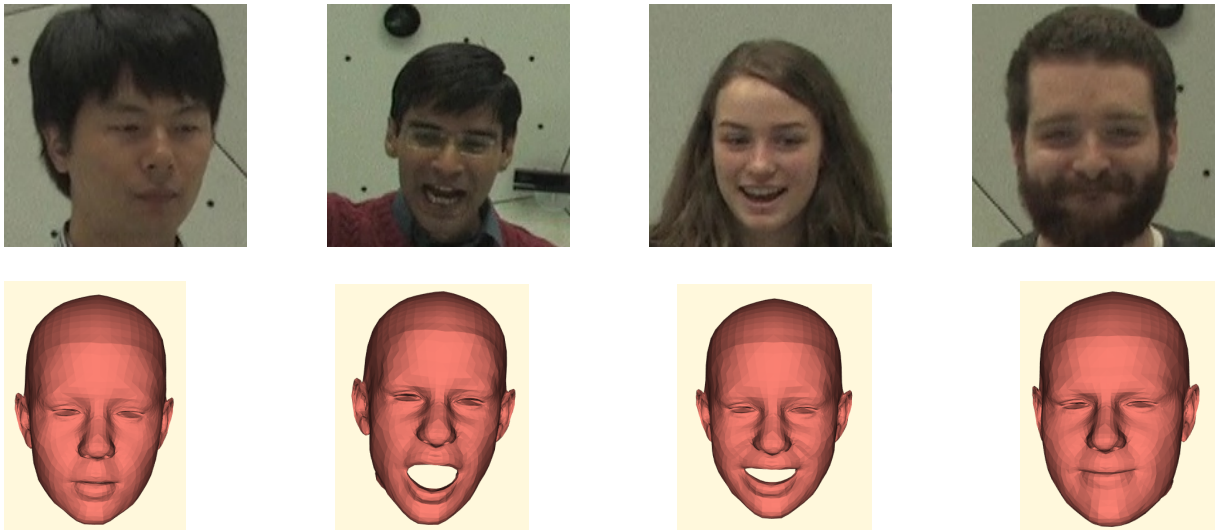


Figure 57: Textured visualization (scenes 3, 9, 10 and 15).

Using this visualization tool, the results of the facial reconstruction can be understood more comprehensively, allowing comparisons between results and drawing conclusions about the associated improvements and errors. It is noteworthy that the shape, size, and

expression of the head generally match the images, although some deformations may appear in certain frames.

5.3. Results of the landmark detectors with the new annotations

After carrying out the training procedures specified in Section 4.4, two neural networks capable of detecting facial landmarks were obtained. The loss curves for both cases are shown in Figure 58. The first corresponds to the training that uses Equation 13 as the loss function, and the second to the training that applies Equation 16 according to landmark visibility. It should be noted that, due to this difference, the training losses cannot be directly compared, as the calculations are not the same. However, the loss used during the validation phase was computed in the same way for both methods, making them comparable.

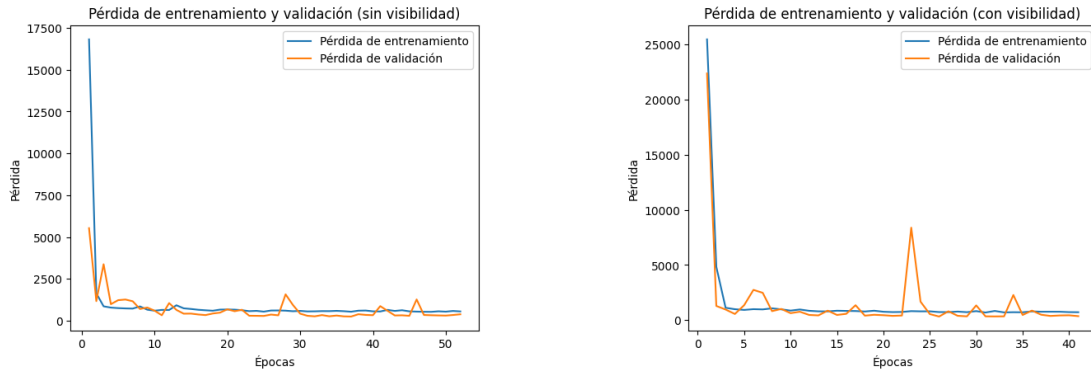


Figure 58: Training and validation losses.

When analyzing Figure 58, it can be observed in both scenarios that the loss is initially high during the first epochs but decreases drastically at the beginning of training. Once this value increases consecutively for 15 epochs, the process is automatically stopped, using this measure to prevent overfitting. In the first case, a total of 52 epochs are performed, reaching a final loss value of 380.29, while in the second case the training completes after 41 epochs with a final value of 353.33. By computing the RMSE through the square root of these values, the measure in pixels can be obtained, resulting in reprojection errors of 19.50 and 18.80, respectively. Nevertheless, although the second training concludes with a higher final loss, the first one achieves a lower minimum loss.

After performing inference on the test images, the mean error between the predictions and the annotations is calculated for both models. For the neural network trained with equal loss for all points, a mean error of 304.42 is obtained, corresponding to 17.44 pixels, whereas the visibility-based loss model yields a mean squared error of 346.84 (18.62 pixels). Therefore, the error remains lower for the model that considers the positional information of all *landmarks* equally.

To facilitate the interpretation of the results, the inferences for some test images are presented in Figure 59. The predictions from the training without visibility consideration are shown in green, the points predicted by the visibility-based loss network are in red, the points provided by DAD-3DNet are in blue, and the refined 3D reconstruction landmarks are shown in white.

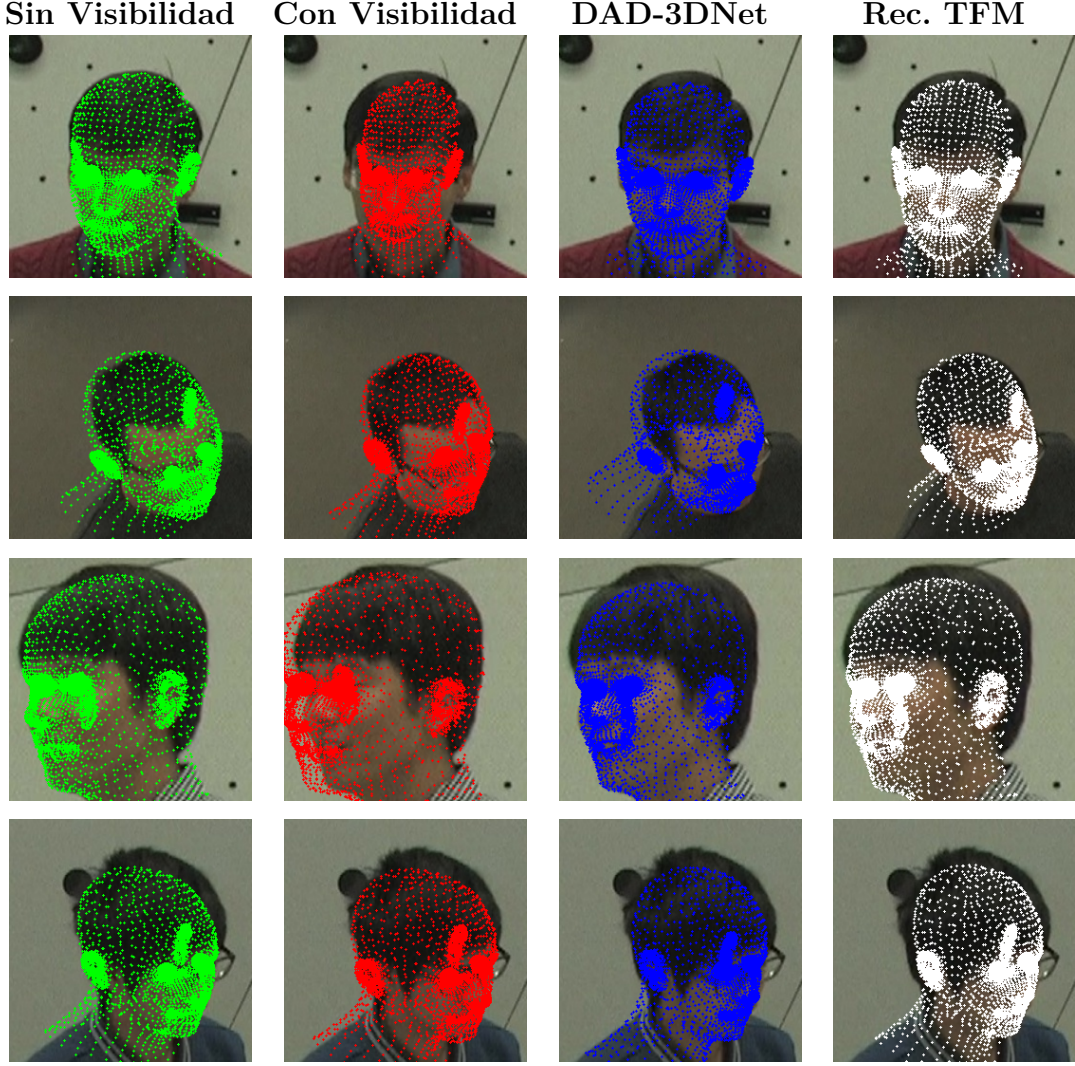


Figure 59: Example of the obtained predictions.

It is worth noting that the *landmark* detector that does not take point visibility into account generally achieves better results in localizing head points compared to the visibility-based detector. In addition, the overall head size is better adjusted than in the presented examples. On the other hand, it can be observed that DAD-3DNet and the refined reconstructions derived from this same neural network exhibit remarkable consistency in their results.

In Figure 60, some examples are shown where the *landmark* detectors encounter difficulties when predicting on the images. In these cases, it can be seen that the models are able to roughly orient the head points, but they fail to properly fit the facial shape for either detector. Despite the fact that the training of these detectors was carried out with similar Panoptic images, both DAD-3DNet and the reconstructions developed in this thesis achieve better approximations. Moreover, this statement is supported by the investment of resources and dedicated research in the development of the DAD-3DNet network, compared to the trained model, which has lower complexity and fewer parameters. It should also be noted that the training of DAD-3DNet was conducted using manually created *ground truth*, whereas the labels used for this training contain errors in some cases, which may lead to reduced performance.

Comparing the training results in general, it can be observed that the inclusion of visibility

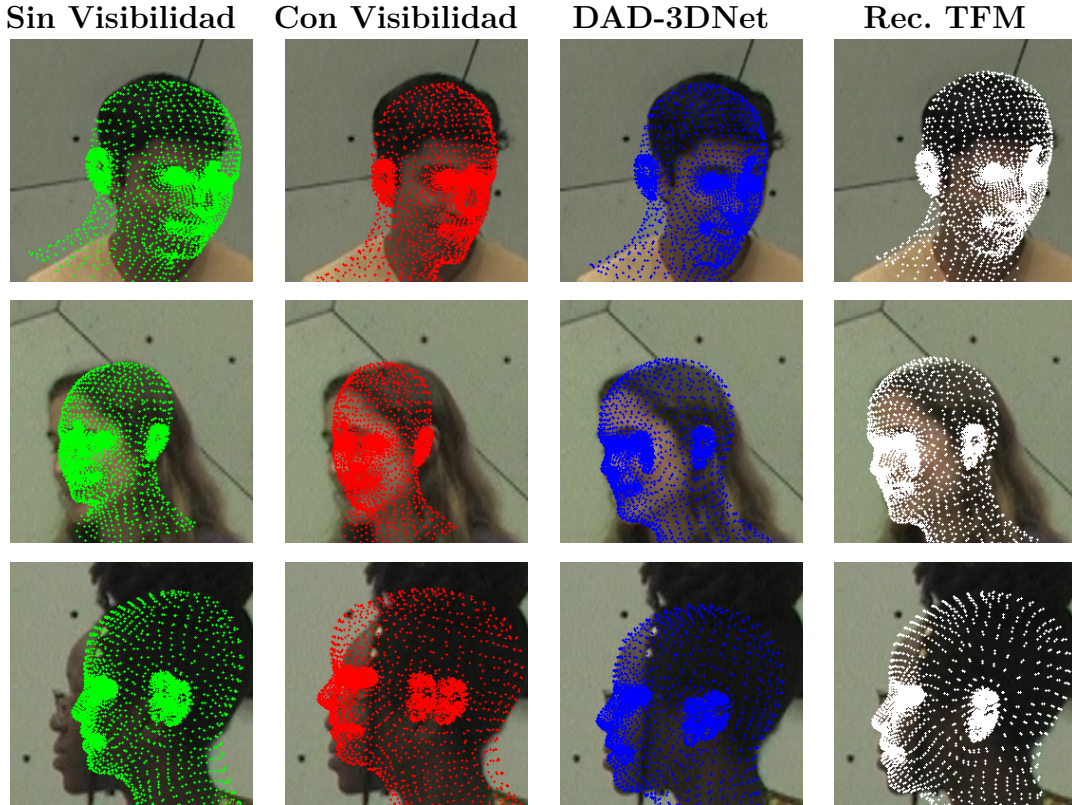


Figure 60: Example of certain errors in the obtained predictions.

information in this process does not provide an advantage compared to using an equal loss for all points. Although one might initially assume that the network with visibility-based loss would better fit the observable part of the image, in practice this premise worsens the quality of the results. By examining the detections obtained by both neural networks, it can be concluded that the model that does not take visibility into account delivers higher-quality results. This is also supported by the lower loss obtained during the validation and test phases, which translates into a better fit of the 5023 points. However, it does not reach the maturity required to be considered an accurate landmark detector. Nevertheless, these results depend on the training process and the number of epochs.

Therefore, it is evident that addressing these shortcomings would require more rigorous training, with a larger number of samples, more complex neural networks, and better filtering of incorrect annotations in order to reach the prediction quality of DAD-3DNet. Even so, it is important to highlight the approach of considering point occlusion and the use of annotated Panoptic scenes, which provides preliminary results regarding their usefulness for the subsequent development of new models.

Despite this, the prediction trends of both models are correct, and it is observed that if the subject's pose is not compromised, some good landmark detections can be obtained. It can be stated that the new annotations can indeed be used in landmark detector training, validating their usefulness and fostering the development of new facial analysis systems.

Chapter 6

Conclusions

As explained throughout the project, 3D facial models can be used in a wide variety of applications, ranging from entertainment, the creation of virtual models, facial analysis for diagnostic purposes, and other approaches based on facial recognition for security. However, for any of these areas, there is a lack of complete databases of 3D reconstructions that can be used in the development of new applications with more accurate models.

The work carried out has been developed with two interrelated purposes in mind. First, to annotate the facial reconstructions in the Panoptic dataset by applying a refinement to the DAD-3DNet predictions, and second, to use these new data for training while taking point occlusions into account.

Through the development of the method for obtaining 3D reconstructions using multiview information, the importance of using high-quality data that can guarantee the necessary accuracy for these types of three-dimensional models has been highlighted. One of the fundamental steps in this development was the initial preprocessing performed on the Panoptic images. In this case, a preliminary filtering was achieved to avoid making predictions on camera images where the subject's face could not be properly seen.

Furthermore, after obtaining the DAD-3DNet predictions on these images, the algorithm for performing linear reconstructions with a different number of cameras was understood and implemented, as well as applying the refinement with [BA](#) through the installation of an external library.

Similarly, different strategies were explored to obtain accurate facial reconstructions, which were compared with the Panoptic annotations and the manually annotated key points in the evaluation scenes. In this context, the impact of using a different arrangement or number of cameras on the final total reconstruction error was studied. To carry out this analysis, three methods were proposed by comparing the quality of their results against a random ordering of the cameras. In this way, it was verified that the two methods based on ordering the cameras according to their error against Panoptic provided the overall results with the lowest error. However, the pairwise method without [RANSAC](#) presents a significant disadvantage if a large number of reconstructions is to be processed and the number of cameras to analyze increases progressively. Its computational cost, together with the required time, as well as a higher error in some cases, makes the [RANSAC](#) method with Panoptic more suitable for this task, even though it may skip some comparisons during the iterations. Using this algorithm, higher-quality data could be obtained by limiting the number of iterations to 50. For this reason, this method was

selected to carry out the final annotations. Likewise, this procedure can also be adjusted to reduce execution time by performing fewer iterations, although the error would increase accordingly. Regarding the [RANSAC](#) algorithm that checks the results against the DAD-3DNet predictions, it is worth noting that it compares all 5023 landmarks, thereby taking the entire head into account when ordering the cameras. Nevertheless, the error obtained for the 27 manually annotated points is higher, and the required execution time is not reduced. Moreover, it was observed that randomly ordering the cameras increased the reprojection error compared to using one of the aforementioned ordering and selection algorithms. These errors become more significant when a very small number of cameras is used. Still, if all cameras are used in the final reconstruction, the final error tends to stabilize across all methods.

It is important to point out that, although the organization and selection of cameras that provide the best predictions on the head reference points can influence the results, in general terms, if a truly fast processing is sought, all cameras could be used in the reconstruction to obtain valid results.

Additionally, the visualization provided by VTK through polygon processing of the points to create textures should also be mentioned. In this way, some errors that could go unnoticed during evaluation could be visualized in a 3D space. Furthermore, this tool is highly effective for exploring results throughout a sequence of frames.

After this initial processing, the annotations created on a set of Panoptic sequences were used to carry out training for obtaining landmarks detector models. To this end, the 3D reconstruction results in XYZ space were reprojected onto each of the images used, obtaining the new refined 2D key points. Although retraining DAD-3DNet was initially considered, it was concluded that the complexity of the neural network and the data exceeded the scope of this project. Therefore, to try to obtain the best possible results using the new data, two different loss functions were proposed.

First, the neural network was trained with a loss in which all facial reference points had the same weight in the error calculation. Then, as a second approach, the training was configured to consider the visibility of these points. To obtain this feature inherent to the head orientation in the images, trigonometric procedures were applied to the 3D reconstruction points, filtering the landmarks visible according to the head orientation in the 2D image. In this way, a custom loss could be applied, focusing on giving more importance to the landmarks visible in the image, in order to improve the global fit according to the visible part of the image. Additionally, to carry out this training, a data update protocol had to be established to obtain this visibility property from the corresponding *batch* during training.

Evaluation of these training techniques revealed certain characteristics in the results. On the one hand, the validation and test error was lower for the training with equal-weight loss for all points. This detector achieved better results than the model trained with the visibility feature, fitting the facial landmarks more accurately in the images. Although the detectors still present some errors, the results show an appropriate trend. Using a simple model and the new annotations, a first approach in the development of detection neural networks was achieved, producing predictions in line with expectations and fulfilling the goal of validating the new annotations. It is worth noting that using the fully annotated database, together with more complex neural networks, could enable higher-quality training and the development of more precise applications.

Therefore, a limiting factor of this work has essentially been the quality of the annotations used to evaluate the refinement of the 3D reconstructions. Both the DAD-3DNet predictions and the Panoptic database annotations affect the final result since the ground-truth used was not manually created. Despite this, the obtained results are promising and align with the main objective of the project, which is to create more accurate databases for the generation of 3D models.

In conclusion, it is important to highlight that the project objectives were achieved thanks to the application of the proposed methodology. In this way, the Panoptic dataset has been successfully annotated using filtering and refinement methods. In this process, the importance of employing 3D reconstruction strategies that allow the fusion of data from different viewpoints, improving the quality of the results, must be emphasized. Furthermore, the trained models are able to make predictions close to the expected outcomes and fulfill the ultimate purpose, which is essentially to apply the refined annotations to new processing tasks.

Chapter 7

Future work

With the aim of continuing the work of this project, certain improvements are proposed in the following areas:

- The reconstructions obtained in this Master’s Thesis depend on the quality of a FLAME parameter estimation system, i.e., a 3DMM model. An improvement could come from refining the DAD-3DNet model (*finetuning*) to use multiview and temporal information from the video in a coordinated manner.
- Although fully manually annotating Panoptic is unfeasible, it would be beneficial to partially annotate this database. This approach could lead to better results in the camera selection algorithm, achieving higher accuracy and reliability in the reconstructions.
- Regarding the training processes, using more complex neural networks together with the full set of processed Panoptic data could enable the development of higher-quality systems that are more robust to different perspectives. With this information, models could be created both for detecting 2D facial landmarks and for generating 3D head reconstructions from a single image. Additionally, incorporating multiview information could be highly beneficial in this type of training.

By incorporating the improvements mentioned, it can be concluded that a significant increase in result quality would be achieved, leading to the generation of more robust and efficient models. These modifications not only aim to optimize quality but also enhance the integration of these annotations into future projects.

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